Beyond Ranks:

Inequality in the Measurement of Mobility*

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Abstract

We argue the change in individuals' Lorenz ordinates—their positions in the Lorenz curve—is informative about economic mobility. Lorenz ordinates combine the ordinal content of ranks with the cardinal content of income differences. In this way, Lorenz mobility is meaningful when there are material differences between incomes across the distribution, explicitly linking inequality and mobility. In a highly unequal society, rank mobility among low-income people produces small income differences, reflected in a relatively flat Lorenz curve, while similar rank movements near the top produce large income differences, reflected in convex Lorenz curves. We show how Lorenz mobility relates to other mobility measures and to standard concepts of income inequality, separating horizontal (or positional) mobility, which holds inequality constant, from vertical mobility, which captures changes in inequality. We also show Lorenz ordinates are axiomatized as individuals' *aspirational* economic status—status increases when an individual's income is closer to those richer than them and further from those at the bottom of the distribution—and provide an application to intergenerational mobility in the US and Norway. Relative to attainable mobility, Lorenz mobility is 10 percentage points is lower than rank mobility, highlighting the role of inequality.

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1. Introduction

The study of mobility and inequality are intertwined, with many measures of economic mobility focusing on the *relative* position of individuals in the economy. These include intergenerational changes in income ranks (Bartholomew 1973 and Shorrocks 1978), and rank correlations (Schiller 1977, Chetty et al. 2014b, or Fagereng, Mogstad, and Rønning 2021).[1] However, purely ordinal measures obscure the connection between mobility and inequality by discarding cardinal information in income. Large changes in ranks may have little material impact when differences in income are small, while small changes in ranks between those with drastically different incomes are meaningful. Without cardinality, relative mobility measures struggle to capture material differences.

The interaction between individuals' positions and cardinal income differences is described by the Lorenz curve—the cumulative income share at each rank. Consider Figure 1A, which shows the Lorenz curve of the log-normal distribution. In more unequal economies, income distributions compress at the bottom, where there is little difference between incomes, and spread at the top, where differences are large. Lorenz ordinates—individuals' positions in the Lorenz curve—capture this relationship: in the presence of inequality, differences in Lorenz ordinates shrink when income differences between ranks are small and spread as these differences increase. We argue this makes the *change in Lorenz ordinates* a desirable measure of economic mobility. They combine ordinal differences between individuals, captured by ranks, with cardinal differences in income levels, reflected in inequality.

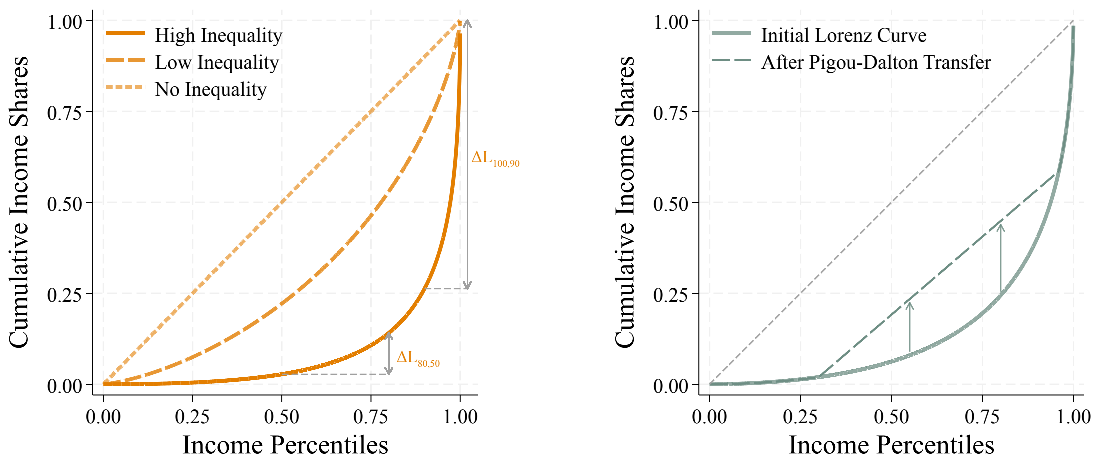
Changes in Lorenz ordinates decompose into *horizontal and vertical mobility* making explicit the implications of inequality for mobility. Horizontal mobility comes from changes in position (or ranks) holding inequality constant: moving along the same

[1] While most of our discussion applies to different economic outcomes, such as wealth, we focus on income throughout for ease of exposition.

FIGURE 1. Inequality and Mobility

(A) Ranks and Lorenz Ordinates

(B) Pigou-Dalton Transfers



Notes: Panel A shows Lorenz curves for log-normal distributions with different variances, equivalently, after successive proportional Pigou-Dalton transfers. The Lorenz curve of a log-normal distribution with variance σ^2 is $L(r) = \Phi\left(\Phi^{-1}(r) - \sigma\right)$; Φ is the standard normal cumulative density. The no-inequality curve is the limit as $\sigma \longrightarrow 0$. Panel B shows a Lorenz curve before and after Pigou-Dalton transfers.

Lorenz curve. These changes are meaningful in as much as inequality makes their Lorenz ordinates different (Figure 1A). Vertical mobility comes from changes in inequality that alter the Lorenz curve. Reducing inequality increases mobility (by increasing Lorenz ordinates), as is the case after Pigou-Dalton transfers from richer to poorer individuals (Figure 1B). When individuals keep their position in the distribution or when aggregating signed changes in Lorenz ordinates mobility only reflects changes in inequality. In this way, our results highlight that mobility is not separable from the distribution of income, linking to extensive work interpreting mobility as social welfare (Atkinson and Bourguignon 1982). In contrast, mobility in income ranks are invariant to changes in monotone changes in the distribution, like those from progressive taxation, ignoring the effects of redistribution.

We show how to aggregate the changes in Lorenz ordinates across the population to obtain measures of aggregate signed mobility (Bhattacharya and Mazumder 2011;

Cowell and Flachaire 2018; Ray and Genicot 2023), symmetric mobility (Fields and Ok 1996, 1999), and intergenerational correlations (Hart 1983).[2] The justification for these measures, such as the axiomatic derivation in Shorrocks (1993) for Hart's mobility measure, applies equally to mobility in Lorenz ordinates. Our approach also disentangles overall economic growth from mobility providing a measure comparable across economies with units expressed in income shares.[3] Moreover, we characterize the maximal Lorenz mobility, allowing us to compare mobility in Lorenz ordinates to rank mobility or other units.

The case of signed mobility is particularly instructive as it provides an explicit link between mobility and inequality. The average signed change of Lorenz ordinates—that is, *net mobility*—corresponds to the change in the Gini coefficient, so that a decrease in inequality corresponds to an increase in mobility regardless of whether Lorenz curves intersect. Effectively, net mobility coincides with average vertical mobility as rank changes (and therefore horizontal mobility) wash out. Crucially, our notion of net mobility is panel independent so it can be readily computed from repeated cross-sectional data or datasets without both generations (as in Ray and Genicot 2023).

Aggregate mobility can itself be decomposed into the product of the average mobility and the concentration of that mobility in the population, that is, how unequal mobility is. The result is a representation of aggregate mobility familiar from information-theoretic concentration measures of inequality (Theil 1967) and those that construct equally-distributed-equivalent social welfare measures (Kolm 1969; Atkinson 1970; Sen 1973). In this way, average mobility speaks to the level of mobility in the economy (shaped

[2]These are commonly implemented by computing the intergenerational elasticity of income (e.g., Solon 1992; Zimmerman 1992; Dearden, Machin, and Reed 1997; Mazumder 2005; or Bolt, French, Hentall-MacCuish, and O'Dea 2024) or the Spearman correlation of income ranks (e.g., Dahl and DeLeire 2008; Olivetti and Paserman 2015; Ward 2023; or Jácome, Kuziemko, and Naidu 2025). Alternative rank-based measures are derived from transition matrices (e.g., Dardanoni 1993; D'Agostino and Dardanoni 2009).

[3]Our Lorenz-based approach also complements studies on top income and wealth shares (see Kopczuk 2015 for a survey). Relatedly, Audoly et al. (2025) use Lorenz ordinates to study wealth mobility as an alternative to ranks and log-wealth that emphasizes top wealth holders.

by the change in inequality) that is then penalized (or augmented) depending on how concentrated (unequal) mobility is. In effect, the mobility concentration provides the equivalent fall in inequality required to exactly offset unequal mobility, completing the link between inequality and mobility.

We apply our framework to intergenerational mobility in the US and Norway using linked parent-child data. Mobility using Lorenz ordinates is lower than mobility in ranks reflecting the role of inequality in both countries. Relative to attainable mobility, rank mobility is 10 percentage points higher than Lorenz mobility in the US and Norway. Moreover, mobility is uneven and concentrated among those making large transitions. Despite similar Lorenz curves across generations (hence low vertical mobility), changes in position more than offset an increase in inequality between generations and generate mobility equivalent to 26–30% of aggregate income.

Finally, we show Lorenz ordinates provide a natural measure of *aspirational* economic status, justifying its use for mobility. This characterization arises under two key conditions. First, status is *aspirational* in that it only depends on the income distance to the top and the bottom of the distribution: individuals *aspire* to catch up to those richer than them without compromising their position relative to those poorer.[4] Second, status satisfies *anonymity* with respect to others: it depends on how much income they hold, not who they are or how income is distributed among them.

The result is that an individual's status is determined not only by their position in the distribution (their rank) but also by the share of income held by those richer (and poorer) than them, increasing when there is less (income) distance to the top (and more distance to the bottom). It is in this sense that transfers from the rich to the poor increase mobility. Receiving transfers increases status by bringing individuals closer to those richer than them (and further away from those at the bottom) even if relative

[4]Economy wide outcomes determine individual aspirations as in Genicot and Ray (2017). However, unlike that paper, we treat aspiration continuously through movements in the income distribution.

positions (ranks) are preserved (Figure 1B).

While not driven by equity concerns, our results share properties with theories of inequality aversion (Robson 1992) and fairness (Fehr and Schmidt 1999) where the position of an individual in society is relevant (see also Lazear and Rosen 1981). However, position is not sufficient as an increase in income is valued even if ranks remain constant, underscoring the role of income for the consumption capabilities of individuals (Sen 1973). Our characterization of individual status via Lorenz ordinates is also linked to indexes of *aggregate satisfaction* in the context of relative deprivation (e.g., Runciman 1966; Yitzhaki 1979; Kakwani 1984), despite the two being conceptually different. Satisfaction indexes aggregate income comparisons between individuals, as the axiomatization in Ebert and Moyes (2000) makes clear, while status captures an individual's capabilities and position in society.

2. Inequality and Mobility

We are interested in mobility beyond ranks, so that cardinal income changes matter *as well as* the position of individuals in the income distribution. We argue for the use of changes in Lorenz ordinates to measure mobility and proceed by establishing their properties and the link they provide between inequality and mobility. We also give a justification for the use of Lorenz ordinates and income ranks as measures of economic *status* in Section 5.

We consider an economy populated by a finite set of dynasties indexed $i = 1, \ldots, N$. There are two observations of income for each dynasty, y^P and y^K, these might be incomes of parents and their children or of the same individual at different dates. We denote by Y^P and Y^K the $(N \times 1)$ vectors of income. We define functions r and L as the rank and the Lorenz curve of Y, respectively,

$$r(y, Y) \equiv \sum_{i=1}^{N} \frac{\mathbb{1}_{\{y_i \leq y\}}}{N} \quad \text{and} \quad L(y, Y) \equiv \sum_{\{\tilde{y} \in Y \mid \tilde{y} \leq y\}} \frac{\tilde{y}}{N\mu}; \quad \text{where} \quad \mu \equiv \sum_{i=1}^{N} \frac{y_i}{N}. \quad (1)$$

When convenient we will refer to $r_i^G = r\left(y_i^G, Y^G\right)$ and $L_i^G = L\left(y_i^G, Y^G\right)$ as the rank and Lorenz ordinate of dynasty i in generation $G \in \{P, K\}$. It is immediate to rewrite the Lorenz ordinates as a function of ranks—the income share of individuals in G with rank lower than or equal to r: $L_i^G = \widetilde{L}\left(r_i^G\right) \equiv \sum_{\{j \mid r_j \leq r_i^G\}} \frac{y_j}{N\mu}$. This leads to the graphical representation in Figure 1.

The key difference between Lorenz ordinates and ranks is that the former are strictly increasing in the dynasty's income (holding the income of others constant) while the latter only increase with income if the ordering of the dynasties changes. The two measures coincide only in perfectly equal economies. So, in unequal economies, Lorenz ordinates preserve the cardinal information of income in the measurement of mobility as the Lorenz curve encodes the relative distribution of income (Atkinson 1970; Sen 1973). This makes the role of inequality in mobility explicit.

As shown in Figure 1A, a change in ranks in an unequal economy translates into a large change in Lorenz ordinates if it corresponds to a meaningful change in income (relative to the average income in the economy). When income differences are small, as between low-income dynasties, small changes in income can generate large changes in ranks, but not meaningful changes in the consumption capabilities of the dynasty. When income differences are large, small changes in ranks can require large changes in income that are reflected in the steepness of the Lorenz curve.

Lorenz ordinates, like ranks, are bounded between 0 and 1 and have clear and interpretable units in terms of *income shares*. Specifically, Lorenz ordinates are the same under incomes $\{Y^P, Y^K\}$ as in an economy where average income is constant across generations, $\left\{\mu^K/\mu^P Y^P,\ Y^K\right\}$; that is, they are *scale invariant* within and between generations as in Shorrocks (1993).

This makes redundant the units in which income is measured and separates overall income growth from mobility, so that Lorenz ordinates capture *relative* as opposed to *absolute* mobility. Consequently, an increase in the Lorenz ordinate by 0.1 for dynasty i is equivalent to the dynasty increasing their own income by 10% of aggregate income (fixed at the income level of either generation).

2.1. Inequality, redistribution, and mobility

Measuring *Lorenz mobility*—the change in Lorenz ordinates—provides a clear link between inequality, redistribution, and mobility. Concretely, mobility is the change in how much of the economy's income lies beneath you—equivalently, how much income lies above you—and movements towards equality increase upward mobility. In this way, Lorenz ordinates capture both the cardinality of income—as they are increasing in the dynasty's income holding the distribution constant—and the role of inequality in the economy—as they are increasing when the distribution becomes less unequal.

Consider two co-monotone income distributions, when generations share the same ranks or when comparing market and disposable income under a monotone tax system (progressive or otherwise). In this case, the position of each dynasty does not change. Consequently, all mobility—between or within generations—is due to changes in inequality.[5] In contrast, rank-based measures of mobility enforce zero mobility in all cases, irrespective of the shapes of the income distributions. *Rank-based measures of mobility are invariant to taxation* when the tax system is monotone, even if the tax

[5]Co-monotonicity only restricts the relationship between ranks, imposing no restrictions on the shape of Lorenz curves which can cross so that some dynasties experience upward mobility while others experience downward mobility. See the discussion of Axiom 5 on page 32.

system changes over time.[6]

More generally, the link between redistribution and mobility is made clear by contrasting the effects of *Pigou-Dalton transfers*—that redistribute income to reduce inequality—with the effects of *income swaps*—that change dynasties' positions without changing inequality.[7] Consider an arbitrary sequence of Pigou-Dalton transfers; each transfer locally reshapes the income distribution to reduce inequality. This necessarily increases Lorenz ordinates in the economy in the sense of the usual stochastic order (Atkinson 1970). The impact on mobility, however, depends on both shifts in the Lorenz curve and positional changes of dynasties after transfers, to which we return shortly in Section 2.2. Yet, compared to a sequence of income swaps that produce the same positional changes without changing inequality, Pigou-Dalton transfers increase each dynasty's mobility. Dynasties who retain or increase their rank experience greater upwards mobility, vice versa those whose ranking decreases, as the Lorenz curve after Pigou-Dalton transfers is pointwise above the original (see Figure 1B).

Proportional Pigou-Dalton transfers, which smoothly reduce inequality across the distribution, illuminate the role of pure redistribution. These transfers deliver a new vector $Y^{\alpha} = \alpha Y + (1-\alpha)\,\mu$ for some $\alpha \in [0,1]$, and redistribute income from dynasties with above-average income to those with below-average income. They do so without changing any dynasty's position in the income distribution: preserving ranks, but raising the Lorenz ordinate of all dynasties (Figure 1A). Formally,

$$L\left(y_i^{\alpha}, Y^{\alpha}\right) = \alpha L\left(y_i, Y\right) + (1-\alpha)\, r\left(y_i, Y\right) \geq L\left(y_i, Y\right) , \tag{2}$$

[6]This includes the intergenerational rank correlation, conditional expected rank for children born to parents at the p$^{\text{th}}$ percentile, $E[r_i^K | r_i^P = p]$, or the probability a child born to parents in the bottom-quintile is in the top-quintile of their own generation's distribution.

[7]Formally, consider two dynasties h and j with $y_h < y_j$. A transfer $\delta > 0$ from j to h is a Pigou-Dalton transfer if j is not poorer than h after the transfer, $y_h + \delta \leq y_j - \delta$. These transfers maintain the relative order of the two dynasties h and j but can induce positional changes with respect to other dynasties, for instance increasing the rank of h and correspondingly decreasing the rank of a third dynasty h'.

where the inequality follows from the fact that $r_i \geq L_i$ for all i. This type of redistribution has the largest impact for dynasties in the middle of the distribution, even when dynasties in the extremes have the largest changes in incomes.

Pure redistribution highlights a tension between purely ordinal measures of mobility, that are insensitive to redistribution, and purely cardinal measures, that do not take the distribution of income into account. Lorenz ordinates resolve this tradeoff based on how inequality changes (something that becomes explicit when measuring average mobility, as we show in Section 3.2). They reflect cardinal changes in income affecting a single dynasty, and evaluate economy-wide changes in income in the light of changes in inequality, effectively measuring the *position* of a dynasty with the *income distance* to the bottom and the top of the distribution. This is why the Lorenz ordinate of a dynasty can increase even as their income remains constant or decreases, provided their income change is accompanied by changes in the distribution of income that "move them up," simultaneously increasing their income distance to the bottom and decreasing their distance to the top of the distribution.[8] In this way, Lorenz ordinates give the *position* of the dynasty *cardinality* content based on inequality.

Market income, disposable income, and mobility. One important implication of the link between mobility and inequality is that Lorenz mobility in market and disposable income will generally differ. This difference is driven by how the Lorenz curve of disposable income incorporates the redistribution enacted by the tax system.[9] In contrast, and as noted above, rank-based measures of mobility in market and disposable income do not differ as they are invariant to taxation.

Differentiating between market and disposable income is relevant for the

[8]Of course, the Lorenz ordinate would decrease if the income of the dynasty decreases in isolation.

[9]This result, while intuitive, is not immediate. This is because taxation does not correspond in general to an application of successive Pigou-Dalton transfers. Nevertheless, the role of tax progressivity is important. A simple linear tax does not affect the shape of the Lorenz curve.

interpretation of mobility. While market income is often used in the study of inequality and mobility as a measure of an individual's consumption capabilities (Piketty, Saez, and Zucman 2018, e.g.,) it is also understood to be an imperfect measure, not least because of the role of the tax system in redistributing resources. This makes disposable (or after-tax) income an informative measure of an individual's available resources and consumption capacity, while market income is potentially more informative about skill and earning ability.

2.2. Disentangling changes in position and changes in inequality

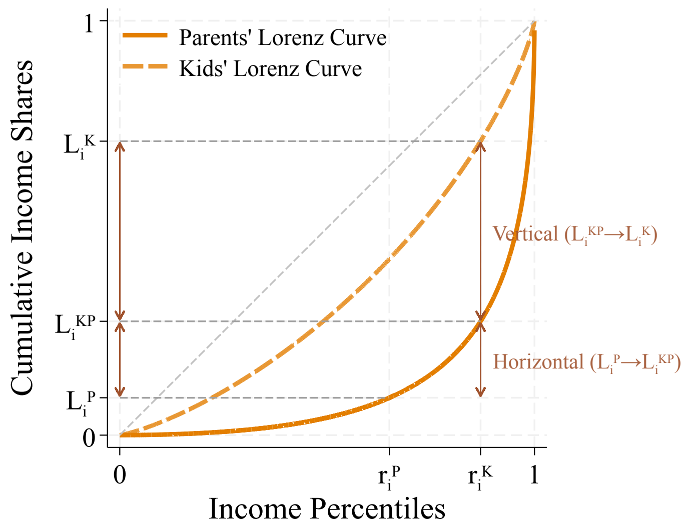
Our discussion emphasizes that Lorenz mobility reflects not only changes in a dynasty's position in society but also changes in inequality over time. We now disentangle these two forces driving mobility. Formally, we distinguish between *horizontal mobility* coming from changes in rank or position under the same income distribution, and *vertical mobility* coming from changes in inequality between the origin and the destination income distributions, while holding rank constant. In this way, Lorenz mobility can be written as

$$L_i^K - L_i^P = \overbrace{L_i^K - L_i^{KP}}^{\text{Vertical Mobility}} + \underbrace{L_i^{KP} - L_i^P}_{\text{Horizontal Mobility}}, \tag{3}$$

where $L_i^{KP} \equiv \widetilde{L}^P\left(r_i^K\right)$ is the Lorenz ordinate of generation K under P's income distribution while retaining the rank they have under K's distribution.[10] The distinction is complete in two polar cases. Mobility between co-monotone generations is entirely vertical—regardless of changes in the income distribution, while mobility between

[10]It is also possible to decompose mobility prioritizing changes in inequality, moving first between Lorenz curves holding constant the parent's rank and then taking into account the change to the kid's rank. This gives $L_i^K - L_i^P = \left(L_i^{PK} - L_i^P\right) + \left(L_i^K - L_i^{PK}\right)$, where the first term is vertical mobility and the second horizontal mobility.

FIGURE 2. Horizontal and Vertical Mobility



Notes: The Figure illustrates horizontal and vertical mobility for a dynasty moving from rank r_i^P in the parents' Lorenz curve (dark solid line) to rank r_i^K in the kids' curve (light dashed line). Horizontal mobility corresponds to the movement from L_i^P to the Lorenz ordinate of an individual with rank r_i^K in the parents' Lorenz curve, L_i^{KP}. Vertical mobility corresponds to the movement from L_i^{KP} to the ordinate L_i^K.

generations with the same distribution of income is entirely horizontal—regardless of changes in positions.

Figure 2 illustrates this. *Horizontal mobility*, the difference between L_i^{KP} and L_i^P, captures mobility coming from rank changes across generations, holding inequality constant. This corresponds to movements along generation P's Lorenz curve. *Vertical mobility*, the difference between L_i^K and L_i^{KP}, captures changes in the distribution of income across generations, holding the dynasty's rank constant. This corresponds to a (vertical) movement from generation P's Lorenz curve to K's curve. In the example, the overall decrease in inequality between generations makes vertical mobility positive for all dynasties. However, Lorenz curves can cross for general changes in the distribution of income, so that some dynasties experience positive vertical mobility while others experience negative vertical mobility.

3. Aggregating mobility

We now turn to aggregating mobility into a single statistic. Formally, we aggregate mobility via a function $\mathcal{M} : \mathbb{R}^N \longrightarrow \mathbb{R}$ from a vector M of dynastic mobilities, with a typical element m_i. We consider two main measures of dynastic mobility: Signed and symmetric mobility, measured respectively by the difference and absolute difference between the Lorenz ordinate of the dynasty in the kid's and parent's generations:

$$\text{Signed mobility:} \qquad m\left(L^P, L^K\right) = L^K - L^P ; \tag{4}$$

$$\text{Symmetric mobility:} \qquad m\left(L^P, L^K\right) = \left|L^K - L^P\right| . \tag{5}$$

The first measure captures cases in which society values the *direction* of mobility, similar to other measures like Ray and Genicot's 2023 upward mobility measure. The second measure accounts for all mobility, regardless of direction, similar to Fields and Ok's 1996 measure for mobility in income levels. We provide conditions on the properties of mobility that lead to these measures in Appendix A.

To aggregate dynastic mobility we adopt the *homogeneous quasi-arithmetic aggregators* axiomatized by the Kolmogorov-Nagumo-de Finetti theorem (see Hardy, Littlewood, and Pólya 1952, Thms. 84, 215). When $M \geq 0$ this gives the standard power mean,

$$\mathcal{M}_N(M) = \left(\frac{1}{N}\sum_{i=1}^{N}\left(m_i\right)^{\gamma}\right)^{\frac{1}{\gamma}} . \tag{6}$$

When mobility is signed, as in (4), the same representation applies with the signed power mean as in Lemma A1 in Appendix A.

These aggregators encompass standard additive aggregators common in the mobility literature (e.g., Bartholomew 1973; Fields and Ok 1996, 1999; Cowell and Flachaire 2018;

Ray and Genicot 2023). They respect monotonicity, so that higher mobility for any dynasty is reflected in higher aggregate mobility, and anonymity, so the indexes of dynasties are superfluous. They also satisfy an *associative* property, so that $\mathcal{M}_N(M) = \mathcal{M}_N([\overline{m}, \ldots, \overline{m}, m_{K+1}, \ldots, m_N])$, where $\overline{m} = \mathcal{M}_K([m_1, \ldots, m_K])$. This property ensures the additivity of the aggregator (and is the reason behind the *quasi-arithmetic* moniker). It implies common properties in the study of inequality and mobility such as replication invariance, so replicating the population $n \in \mathbb{N}$ times does not change aggregate mobility, $\mathcal{M}_{nN}\left(\widehat{M}\right) = \mathcal{M}_N(M)$ for $\widehat{M} = \mathbb{1}_n \otimes M$. We reproduce the formal statement of these results in Appendix A.

3.1. Aggregating unequal mobility

The curvature of the aggregator, γ, determines the emphasis on broad-based versus concentrated mobility in just the same way as in measures of inequality aversion (Atkinson 1970) or choice under uncertainty (as $\mathcal{M}$ also gives the family of certainty equivalents over $\{m_i\}$ under constant relative risk aversion, Ackerberg, Hirano, and Shahriar 2017). When $\gamma = 1$ we recover the average mobility, which neither favors nor penalizes dispersion in mobility. For positive γ, lower values favor broad-based mobility while higher values favor dispersion, effectively placing a higher weight on dynasties with the highest mobility. Negative values of γ place more weight on the dynasties with the lowest mobility.[11]

We can better see the role of γ by separating the roles of *average mobility* and *mobility concentration*. Specifically, aggregate mobility corresponds to average mobility corrected by mobility concentration as in Atkinson (1970). Average mobility corresponds naturally to $\overline{m} \equiv \frac{1}{N}\sum_{i=1}^{N} m_i$. It can be broken up in general into the contributions of upward and

[11]Although γ can tilt aggregate mobility towards the highest or lowest mobility, our representation of aggregate mobility precludes weighting who moves as there are no weights assigned to dynasties based on their identity or initial conditions.

downward mobility

$$\mathcal{U} \equiv \frac{1}{N}\sum_{i=1}^{N} |m_i|^+ , \quad \text{and} \quad \mathcal{D} \equiv \frac{1}{N}\sum_{i=1}^{N} |m_i|^- , \quad \text{with} \quad \overline{m} = \mathcal{U} - \mathcal{D} . \tag{7}$$

When measuring symmetric mobility this reduces to $\overline{m} = \mathcal{U}$, as $\left| m_i^a \right|^- = 0$ for all i.

We measure mobility concentration, $\mathcal{C}$, within upward and downward mobility. Upward mobility concentration is

$$\mathcal{C}_\gamma^+ (M) \equiv \exp\left(\frac{\gamma - 1}{\gamma} D_\gamma^+ (M) \right), \tag{8}$$

where $D_\gamma^+ (M) = \log N - H_\gamma^+(r)$ is the Rényi divergence of the relative upward mobility shares $|r_i|^+ \equiv \frac{|m_i|^+}{N\mathcal{U}} \geq 0$ given by $H_\gamma^+(r) = \frac{1}{1-\gamma} \log\left(\sum_{i=1}^N \left(|r_i|^+\right)^\gamma \right)$. This is a general measure of dispersion allowing for curvature. It builds on the use of information-theoretic concentration measures for inequality (Theil 1967) and follows the equally-distributed-equivalent tradition of Kolm (1969), Atkinson (1970), and Sen (1973).

The weight on mobility concentration depends naturally on the value of γ. When aggregating mobility uniformly, setting $\gamma = 1$, the concentration of mobility does not matter and aggregate mobility corresponds to average mobility. Lower values of γ penalize concentration, reducing $\mathcal{C}_\gamma$, and higher values of γ favor concentration, increasing $\mathcal{C}_\gamma$. Downward mobility concentration is defined analogously and we set it to 0 when measuring symmetric mobility.

The following Lemma formalizes the roles of average mobility and mobility concentration.

LEMMA 1 (**Average mobility and mobility concentration**). *Let* $\overline{m} \equiv \frac{1}{N}\sum_{i=1}^N m_i \neq 0$.

Aggregate mobility under (6) satisfies:

$$\mathcal{M}_N(M) = \underbrace{\overline{m}}_{\textit{Average Mobility}} \times \overbrace{\frac{sign(\mathcal{S}_\gamma)}{sign(\overline{m})}}^{\gamma-\textit{sign correction}} \times \underbrace{\overline{\mathcal{C}}(M)}_{\textit{Net Mobility Concentration}}, \tag{9}$$

where $\mathcal{S}_\gamma(M) \equiv \sum_{i=1}^{N} sign(m_i)|m_i|^\gamma$ *is the signed aggregate of mobility as in Lemma A1 and*

$$\overline{\mathcal{C}}(M) \equiv \left|\left(\frac{\mathcal{U}}{|\overline{m}|}\right)^\gamma (\mathcal{C}_\gamma^+(M))^\gamma - \left(\frac{\mathcal{D}}{|\overline{m}|}\right)^\gamma (\mathcal{C}_\gamma^-(M))^\gamma\right|^{\frac{1}{\gamma}} \tag{10}$$

is net mobility concentration, with upward and downward mobility as in (7) and upward and downward mobility concentration as in (8).

Proof. The result follows from manipulation of the aggregator in (6). □

This collapses to the product of average mobility and overall mobility concentration when $M \geq 0$:

$$\mathcal{M}_N(M) = \underbrace{\overline{m}}_{\text{Average Mobility}} \times \underbrace{\exp\left(\frac{\gamma-1}{\gamma}D_\gamma(M)\right)}_{\text{Mobility Concentration}}, \tag{11}$$

The first term aggregates mobility linearly and, as we show shortly, provides a direct link between changes in inequality and mobility. The second term captures how concentrated mobility is between dynasties, separating the roles of upward and downward mobility when aggregating signed mobility. When mobility is signed, the middle term corrects the sign of aggregate mobility as the curvature γ can make $\mathcal{S}_\gamma$ have a different sign to $N\overline{m} = \mathcal{S}_1$.

3.2. Average mobility and inequality

The average mobility term in (9) links mobility directly to changes in inequality. This is more transparent when aggregating signed mobility (as in 4). Then, average mobility corresponds to the *net* change in dynastic mobility across the economy that is given by the change in the Gini coefficient between generations. So, *net mobility* captures exactly changes in inequality over time.

COROLLARY 1 (**Net Mobility**). *Under dynastic mobility* (4)*, average mobility satisfies:*

$$\overline{m}^s \equiv \frac{1}{N}\sum_{i=1}^{N} m_i^s = \frac{1}{2}\left(\mathcal{G}\left(Y^P\right) - \mathcal{G}\left(Y^K\right)\right), \tag{12}$$

where $\mathcal{G}(Y) = 1 - \frac{2}{N}\sum_{i=1}^{N} L_i$ *is the Gini coefficient.*

When inequality decreases, so $\mathcal{G}\left(Y^K\right) < \mathcal{G}\left(Y^P\right)$, net mobility is positive. A decrease in inequality increases the Lorenz ordinates of dynasties on net. For instance, implementing proportional Pigou-Dalton transfers lowers the Gini coefficient and unequivocally increases mobility as it increases the Lorenz curve of all dynasties (see equation 2 and Figure 1A).[12]

This result holds in general, even under arbitrary Pigou-Dalton transfers when dynasties are subject to both vertical and horizontal mobility as in (3). Net mobility is entirely captured by changes in the Gini coefficient or vertical mobility (as horizontal mobility is zero-sum). In addition, when $\gamma \neq 1$, the net mobility concentration term in equation 9 can be interpreted as the equivalent fall in inequality required to exactly offset unequal mobility. This completes the link between mobility and inequality even when the Lorenz curves of Y^K and Y^P intersect. See Atkinson (1970) and Aaberge and Mogstad (2011) for alternative discussions of ordering intersecting Lorenz curves.

Moreover, the characterization of net mobility in terms of change in the Gini coefficient means that average mobility is panel independent (or aggregate mobility under $\gamma = 1$). It can therefore be measured from repeated cross-sectional data or datasets without inter-generational links (as in Ray and Genicot 2023). This greatly

[12]This result generalizes when aggregating *status mobility* as defined in Section 5. Status depends on weighted Lorenz ordinates and ranks so that net mobility corresponds to Weymark's 1981 Generalized-Gini coefficient, $\widetilde{\mathcal{G}}(Y) \equiv 1 - \frac{2}{N}\sum_{i=1}^{N}\Omega(r_i) L_i$ for positive weights $\Omega(\cdot)$. See Lemma A2 in Appendix A.

facilitates the measurement of aggregate mobility as most data sources on income lack the longitudinal dimension required to follow dynasties (or individuals) over time.[13]

Average *symmetric mobility* is also directly linked to the change in inequality, as captured by the Gini coefficient. We show in Corollary A1 that symmetric mobility is equal to the absolute value of net mobility, adjusted by either downward or upward Lorenz mobility depending on whether overall inequality has increased or decreased.[14] As with net mobility, symmetric mobility increases when inequality decreases, because lower inequality is reflected in an overall increase in Lorenz ordinates. However, symmetric mobility is also shaped by horizontal mobility. Average symmetric mobility therefore reflects gains and losses in Lorenz ordinates arising from dynasties' positional changes, as well as gains or losses arising from changes in the shape of the Lorenz curve that are not summarized by the Gini.

An analytic example with progressive taxation. We illustrate the implications of inequality for average mobility by discussing how it responds to taxation. Consider the case where market income in each generation is Pareto-distributed with tail parameter $\alpha^G > 1$, $G \in \{P, K\}$, so that the Lorenz curve and the Gini coefficient become

$$\widetilde{L}^G(r) = 1 - (1-r)^{1-\frac{1}{\alpha^G}}, \qquad \text{and} \qquad \mathcal{G}\left(Y^G\right) = \frac{1}{2\alpha_G - 1}. \tag{13}$$

We further assume disposable income, $\widehat{y}$, satisfies $\widehat{y}^G = \lambda^G y^{1-\tau^G}$ as in Feldstein (1969); Bénabou (2002); Heathcote, Storesletten, and Violante (2014), where $\lambda^G > 0$ captures the average tax rate and $\tau^G \in [0, 1)$ controls the progressivity, or how quickly average tax rates rise with income. In this case, disposable income is also Pareto-distributed with a

[13] This also makes it possible to construct aggregate mobility from cross-sectional information given assumptions on the concentration of mobility or, more explicitly, on the dependence of ranks between generations, as in Chetty, Grusky, Hell, Hendren, Manduca, and Narang (2017); Berman (2022); Manduca, Hell, Adermon, Blanden, Bratberg, Gielen, van Kippersluis, Lee, Machin, Munk, Nybom, Ostrovsky, Rahman, and Sirniö (2024).

[14] This result parallels the decomposition of income-level mobility in Fields and Ok (1996, Prop. 4.1).

tail parameter $\alpha^G/(1-\tau^G)$.

The effect of taxation on mobility is immediate in the case of net mobility where mobility is given by the change in the Gini coefficient (Corollary 1). A more progressive tax system—higher τ—makes the tail thinner as it reduces inequality. Net mobility reflects changes in inequality and progressivity so that it is positive when $\frac{\alpha^K}{\alpha^P} > \frac{1-\tau^K}{1-\tau^P}$. In general, an increase in progressivity in the children's generation increases net mobility while more progressivity in the parents' generation decreases it. Moreover, tax progressivity affects mobility even when it is constant over time by attenuating differences in income that result in an attenuation of overall mobility.[15] We formalize these results in Lemma A3 in Appendix A.2.

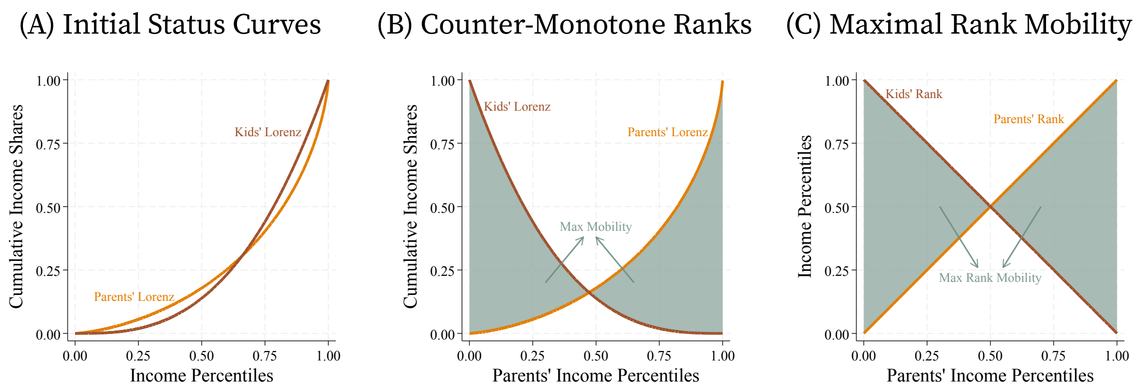
The implications for average symmetric mobility (Corollary A1) are more subtle as it depends not only on the overall contraction in inequality (vertical mobility), but also who is paired with whom across generations (horizontal mobility). Consequently, whether mobility is increasing depends on the degree of intergenerational dependence: as dependence strengthens, the horizontal-mobility component of aggregate mobility fades and the effect of progressivity on reducing vertical mobility dominates. We formalize this in Lemma A4 in Appendix A.2.

3.3. Maximal average mobility

We now derive the maximal average mobility that allows us to express realized mobility as a fraction of attainable mobility, facilitating comparisons across mobility measures. This is immediate in the case of net mobility, as in Corollary 1, for which maximal mobility corresponds to eliminating complete inequality, for a bound of 1/2. If there is information about one of the generations the appropriate bound is obtained as going

[15]The effect of progressivity on Lorenz mobility differs from the effect on measures of income-mobility like the intergenerational income elasticity (IGE). The IGE depends only on relative progressivity across generations so that common progressivity leaves the IGE unchanged. This is likely to occur in applications where the income of both generations is measured in the same time period. See Appendix A.2.

FIGURE 3. Maximal Symmetric Mobility



Notes: Panel A shows Lorenz curves for parents and kids. Panel B shows the maximal average symmetric mobility given these curves obtained by the counter-monotone assignment of parents to kids. Panel C shows the maximal average symmetric rank mobility. Average mobility corresponds to the shaded areas.

from an income distribution with Gini coefficient $\mathcal{G}\left(Y^P\right)$ to one with a Gini coefficient of 0, or going from full inequality to an income distribution with Gini coefficient $\mathcal{G}\left(Y^K\right)$.

When measuring symmetric mobility, as in (5), we can also provide a tight upper bound for aggregate mobility. Given two income distributions (Figure 3A), the highest mobility is achieved when dynasties are ordered in counter-monotone fashion (Figure 3B)[16] because pair-wise mobility is represented by a Monge matrix—a sub-modular discrete function. This result holds regardless of the shape of the status functions and covers horizontal mobility, when the two income distributions are identical, and rank mobility, as in Figure 3C, see Burkard, Dell'Amico, and Martello (2012, Prop. 5.7).

We obtain a global upper bound by maximizing counter-monotone mobility over the space of Lorenz curves. This is achieved when Lorenz curves are extremal in the sense of Baillo, Carcamo, and Mora-Corral (2022). The upper-bound on *Lorenz mobility*

[16]The dynasty with the i^{th} highest income in the parent's generation has the $(N+1-i)^{\text{th}}$ highest income in the children's generation.

when $\gamma = 1$ is obtained as[17]

$$0 \leq \mathcal{M}(M) \leq 2 - \sqrt{2} \,. \tag{14}$$

Maximal *rank mobility* is 1/2 and corresponds to maximal exchange mobility (Bartholomew 1973). See Appendix A.1 for details.

3.4. Intergenerational Lorenz correlations

An alternative approach to measuring mobility estimates the correlation of individual characteristics across generations. Typically, researchers use intergenerational elasticities of income and the Spearman correlation of income ranks, often obtained from *Galtonian* regressions. The coefficient from log-income regressions can be derived axiomatically as a Hart 1983 mobility measure, as shown in Shorrocks (1993), or as the reduced form of Becker and Tomes (1979) models.

Our approach provides a justification for measuring intergenerational correlations of Lorenz ordinates, instead of income or income ranks. The result is an extension of our analysis to the measurement of the intergenerational correlation or elasticity of status.[18] For example, computing $1 - \text{corr}\left(L_i^P, L_i^K\right)$ or estimating

$$\log\left(L_i^K\right) = \beta_0 + \beta_1 \log\left(L_i^P\right) + u_i \,, \tag{15}$$

where the mobility measure is $1 - \hat{\beta}_1$.

[17]Interestingly, this bound is achievable by horizontal mobility. So, it does not require changes in inequality, instead reflecting how inequality shapes changes in status across positions.

[18]The term $\log\left(L\left(y_i, Y\right)\right)$ is well-defined as long as $Y > 0$; the same condition required by $\log\left(y_i\right)$. This approach is also motivated as the best linear prediction of status as measured by Lorenz ordinates.

4. Measuring Lorenz and Rank Mobility

We now apply our framework to studying the intergenerational mobility of Americans and Norwegians. Mobility is higher in Norway than in US, partly reflecting a decrease in inequality between generations in Norway and an increase in the US. Moreover, we find Lorenz mobility is lower than rank mobility in both countries, despite differences in their levels and changes in inequality. Compared to their respective maximal levels, Lorenz mobility is over 10 percent lower than rank mobility.

US Data. We follow Davis and Mazumder (2024) and use data from the National Longitudinal Surveys of Older Men and Young Men and Mature Women and Young Women (NLS66) and the National Longitudinal Survey of Youth 1979 (NLSY79). We exploit a feature of the initial sample design, where multiple respondents were surveyed within the same household, to link parents with children. As Davis and Mazumder demonstrate, this allows researchers to use detailed longitudinal income information on both generations.

We pool the different birth cohorts (approximately those born in the early 1950s and the early 1960s) and weight our analysis using the child's first-round survey weights. The remainder of our sample selection follows Davis and Mazumder (2024) and we refer the interested reader there.[19] Our measure of income for each generation corresponds to total family income averaged over three years.

Norwegian Data. We use detailed longitudinal data on individual income collected by the Norwegian tax authority between 1993 and 2017 as well as demographic information available in their population files. We focus on individual market income from wages and capital. See Blundell, Graber, and Mogstad (2015) for details on the Norwegian

[19] We are indebted to the authors for providing a clear and transparent replication package which allows us to reconstruct their analysis data.

income tax records.

There are several key dimensions in which the Norwegian administrative data complements the survey data we use for the US. The data covers the entire population, including the richest Norwegians, allowing us to construct accurate income ranks and Lorenz ordinates. Second, income is third-party reported, eliminating concerns about measurement error from self-reporting and censoring. Finally, we link incomes across generations without relying on imputation (e.g., Collins and Wanamaker 2022; Ward 2023; or Jácome, Kuziemko, and Naidu 2025) or bounding exercises (e.g., Chetty et al. 2017; Berman 2022; or Manduca et al. 2024).

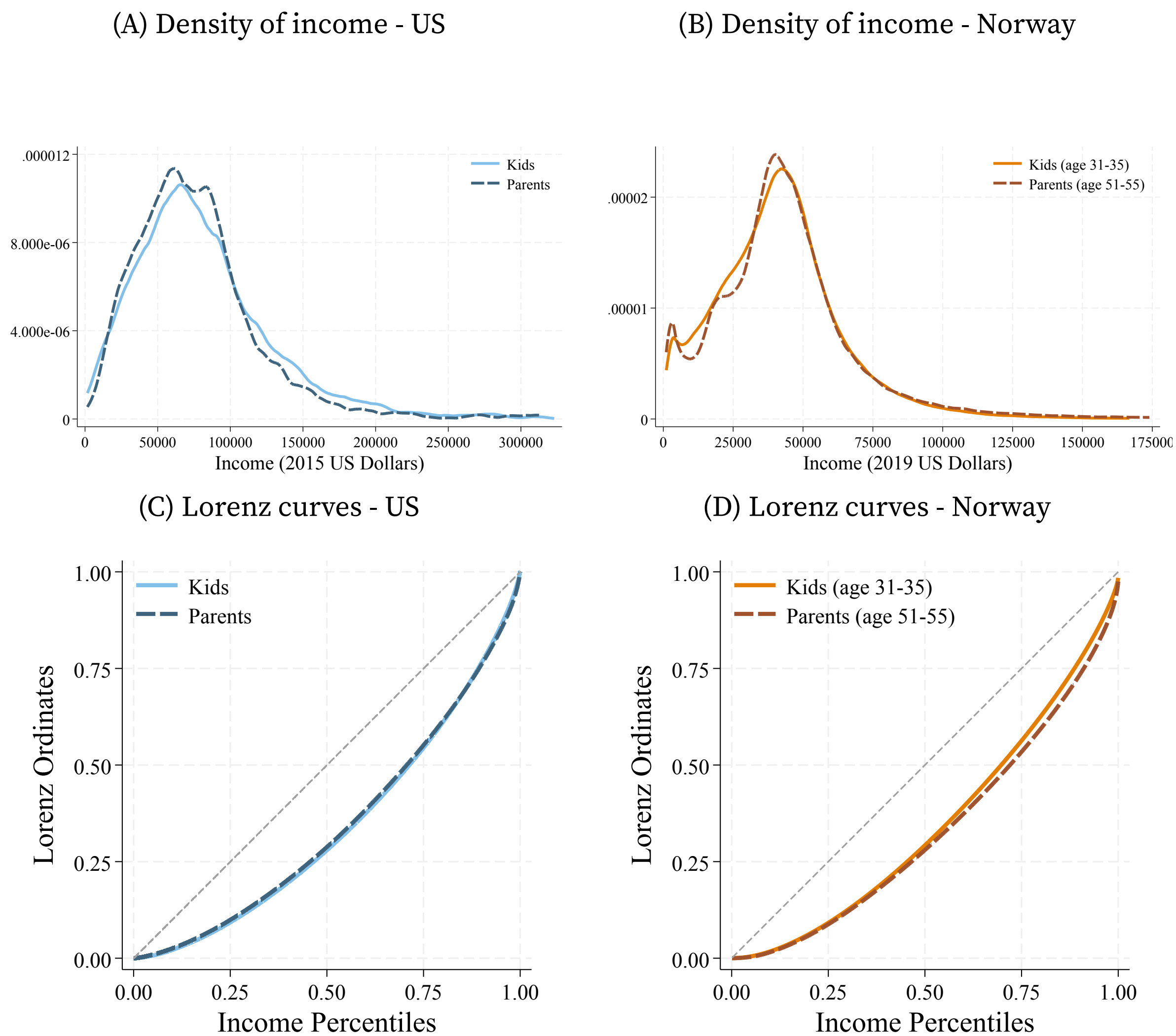
We study Norwegians born between 1961 and 1970 and record their average income between the ages of 31 and 35. We drop individuals who earn less than $1,000 USD before taxes and transfers on average (all dollar amounts are measured in 2019 dollars). We link these Norwegians to their fathers and mothers and record the maximum of their parents' average income between the ages of 51 and 55. This leaves us with 303,451 paired observations.

4.1. Income distribution across generations

The distribution of US children moves rightward relative to their parents, reflecting overall income growth of 7.8%, and has less mass concentrated at low incomes and more mass in incomes above 100,000 dollars (Figure 4A). The reverse happens in Norway, where the average income of children is 7.6% lower than their parents' who also have less mass concentrated at low incomes (except at the very bottom) and more mass in incomes above 100,000 dollars (Figure 4B).

The differences in income distributions translate into small differences in the Lorenz curves between generations. For the US, the Lorenz curves of children and parents are markedly similar, despite crossing at the top of the income distribution (Figure 4C).

FIGURE 4. Income distribution across generations



Notes: Panels A and B show kernel density estimates of income for children and parents in the US and Norway, respectively. Panels C and D show the Lorenz curves for the corresponding income distributions.

The differences are more visible for Norway. The Lorenz curve of children's income is above that of parental income throughout the distribution, evidencing a lower degree in inequality. These differences become more marked above the median.

4.2. Lorenz and rank mobility

We now measure aggregate Lorenz and rank mobility in our two samples. We consider both signed and symmetric mobility (as in equations 4 and 5) aggregated using the homogeneous aggregator in (6) under different curvatures. Table 1 presents the results.

Signed mobility. Average signed mobility ($\gamma = 1$) reflects the small changes in inequality in both samples. In the US, where inequality increases, the Gini rises by half a percentage point between generations and mobility is negative. In contrast, mobility is positive in Norway where the Gini falls by 1.6 percentage points. Ranks capture only horizontal mobility and therefore their aggregate signed changes are 0 by construction.

Mobility itself is also unequal. We see this by comparing our main measure of aggregate mobility $\mathcal{M}$ without curvature, $\gamma = 1$, to measures with lower and higher curvature. When we set $\gamma = 1/2$, aggregate mobility favors a more even distribution of mobility, reducing $\mathcal{M}$ if mobility is concentrated among few dynasties. Indeed, Lorenz mobility is reduced in both samples to essentially zero. Rank mobility is similarly concentrated and remains negligible with $\gamma = 1/2$.

Conversely, $\mathcal{M}$ favors an unequal distribution of mobility when setting $\gamma = 2$, placing more weight on dynasties with higher mobility . For the US, this highlights the presence of large downward Lorenz and rank mobility, both of –5 percentage points on average. For Norway, Lorenz and rank mobility go in different directions. While positional mobility is dominated by large downward moves (3.8 percentage points on average), the decrease in inequality gives large upward Lorenz mobility (over 9 percentage points on average). This is consistent with Norway's more compressed income distribution, which results in more rank displacement that nevertheless has little impact on mobility.

TABLE 1. Aggregate mobility

		Aggregate Mobility $\mathcal{M} = \left(\frac{1}{N}\sum(m_i)^\gamma\right)^{\frac{1}{\gamma}}$					
		Signed Mobility			Symmetric Mobility		
		$\gamma = 1$	$\gamma = 1/2$	$\gamma = 2$	$\gamma = 1$	$\gamma = 1/2$	$\gamma = 2$
USA	Lorenz	–0.005	–0.000	–0.051	0.260	0.212	0.335
	Ranks	—	0.000	–0.052	0.278	0.233	0.349
Norway	Lorenz	0.016	0.001	0.092	0.275	0.227	0.351
	Ranks	—	0	–0.038	0.299	0.251	0.375

Notes: The table reports aggregate signed and symmetric mobility in Lorenz ordinates and income ranks for the US and Norway. The US estimates use total family income averaged over three years; the Norwegian estimates use individual market income. Signed mobility is the change in a dynasty's Lorenz ordinates as in (4), while symmetric mobility is the magnitude of that change as in (5). Aggregate signed mobility is computed using the signed power mean in Lemma A1. Setting $\gamma = 1$ gives average mobility; $\gamma = 1/2$ places more weight on broadly shared mobility; $\gamma = 2$ places more weight on larger movements.

Symmetric mobility. We find symmetric Lorenz mobility is markedly lower than rank mobility, reflecting the role of income inequality in shaping differences across generations in both the US and Norway. Measured relative to their maximal values, Lorenz mobility is over 10 percentage points lower than rank mobility, even as the levels of mobility vary between countries.

For the US, average Lorenz mobility is 0.26, or 44% of its maximal attainable value, while average rank mobility is 0.28, or 56% of its maximal value. In this case, both Lorenz and rank mobility primarily reflect horizontal mobility (as the Lorenz curves of the two generations are close together), but weight positional moves differently reflecting the role of income inequality. The same pattern is present in Norway where mobility is higher. Average Lorenz mobility is 0.28, or 47% of its maximal value, reflecting Norway's decrease in inequality across generations. Rank mobility is 0.3 or 60% of its maximal value. Despite similar values, there are differences in the drivers of mobility—vertical mobility plays a larger role in Norway as Lorenz curves differ across generations.

Symmetric mobility is also concentrated, but not as much as signed mobility. Penalizing concentration (setting $\gamma = 1/2$) reduces Lorenz and rank mobility by 16–18% in both the US and Norway. Placing more weight on high-mobility dynasties (setting $\gamma = 2$) increases measured rank mobility by 26% and Lorenz mobility by close to 30%. Moreover, Lorenz mobility remains below rank mobility under all three values of γ, so accounting for mobility concentration does not alter their ordering.

Disposable income mobility in Norway. The Norwegian data allows us to revisit the relationship between mobility in market and disposable income. We take advantage of the fact that the data is collected by the Norwegian Tax Authority (*Skatteetaten*), providing accurate measures of levied taxes at the individual level, thus we capture effective taxation. Redistribution through taxes and transfers adds to the decrease in inequality already present in market income. This doubles signed Lorenz mobility, from 0.016 for market income to 0.031 for disposable income, as we show in Table A.1 in Appendix A.3. In contrast, symmetric Lorenz mobility is largely unaffected. Thus, on net, the tax system produces additional upward mobility without materially changing the overall movement in Lorenz ordinates. This is consistent with the analytical example in Section 3.2: taxation directly changes mobility, summarized by signed mobility, while symmetric mobility also depends on the horizontal reshuffling of dynasties across generations.[20]

Women's and men's mobility in Norway. We also reproduce our analysis for Norwegian women and men, measuring mobility between mothers and daughters and fathers and sons separately. We present these results in Appendix A.3. Our results for mothers and

[20]Rank mobility is essentially unchanged. Minor differences arise because transfers can break the monotonicity of the tax system, so ranks under market and disposable income need not coincide exactly.

TABLE 2. Intergenerational persistence by country and status measure

	Intergenerational Persistence			
	$\rho\left(x_i^P, x_i^K\right)$			
	Lorenz	Rank	Log-Lorenz	Log-income
USA	0.258	0.229	0.272	0.299
Norway	0.180	0.156	0.100	0.113

Notes: The table reports intergenerational persistence coefficients obtained from regressions as in (15). The Lorenz and Rank columns use Lorenz ordinates and income ranks in levels; the Log-Lorenz and Log-income columns use their logarithms. Higher persistence indicates lower mobility.

daughters echo our previous findings: measured mobility is almost entirely horizontal, as inequality stayed roughly constant across generations despite significant income growth of 18% between generations. Income inequality decreases between mothers and daughters by 0.5 percentage points (a similar magnitude to the increase in inequality in the US). In contrast, we find an important role for decreased inequality among men driving vertical mobility, with a decrease of 3.3 percentage points in the Gini coefficient. Nevertheless, there is more horizontal mobility among women, reflected in larger symmetric mobility than among men. In both cases we verify that Lorenz mobility is lower than rank mobility.

4.3. Intergenerational correlations

Finally, we examine standard measures of intergenerational correlations for log-income, ranks, and status levels. We report these in Table 2 and the corresponding scatter plots in Figure 5. Outcomes are more strongly correlated across generations in the US than in Norway under every measure, consistent with the lower levels of aggregate mobility in Table 1. The correlation of Lorenz ordinates is 0.258 in the US and 0.180 in Norway, while the corresponding rank correlations are 0.229 and 0.156. The same pattern applies when comparing the intergenerational elasticity of income and the correlation of log-Lorenz ordinates. Within each country, Lorenz ordinates are more correlated across generations than ranks, by 0.029 in the US and 0.024 in Norway, again consistent with our previous result of Lorenz mobility being lower than rank mobility.[21] As before, this reflects churn in ranks between dynasties with similar incomes: these positional changes can be sizable in rank units even though they produce small changes in Lorenz ordinates.

5. Income Ranks and Lorenz Ordinates as Economic Status

Finally, we provide a justification for the use of Lorenz ordinates in the study of mobility. Specifically, we provide conditions under which Lorenz ordinates capture the *status* of individuals in society. We see status as capturing the consumption capabilities of individuals (e.g., Sen 1973; Yitzhaki 1979) as well as their position in the economy. Thus, we assume status is a function of an individual's income and the income distribution in the economy, $s : \mathbb{R}_+ \times \mathbb{R}_+^N \longrightarrow \mathbb{R}_+$.

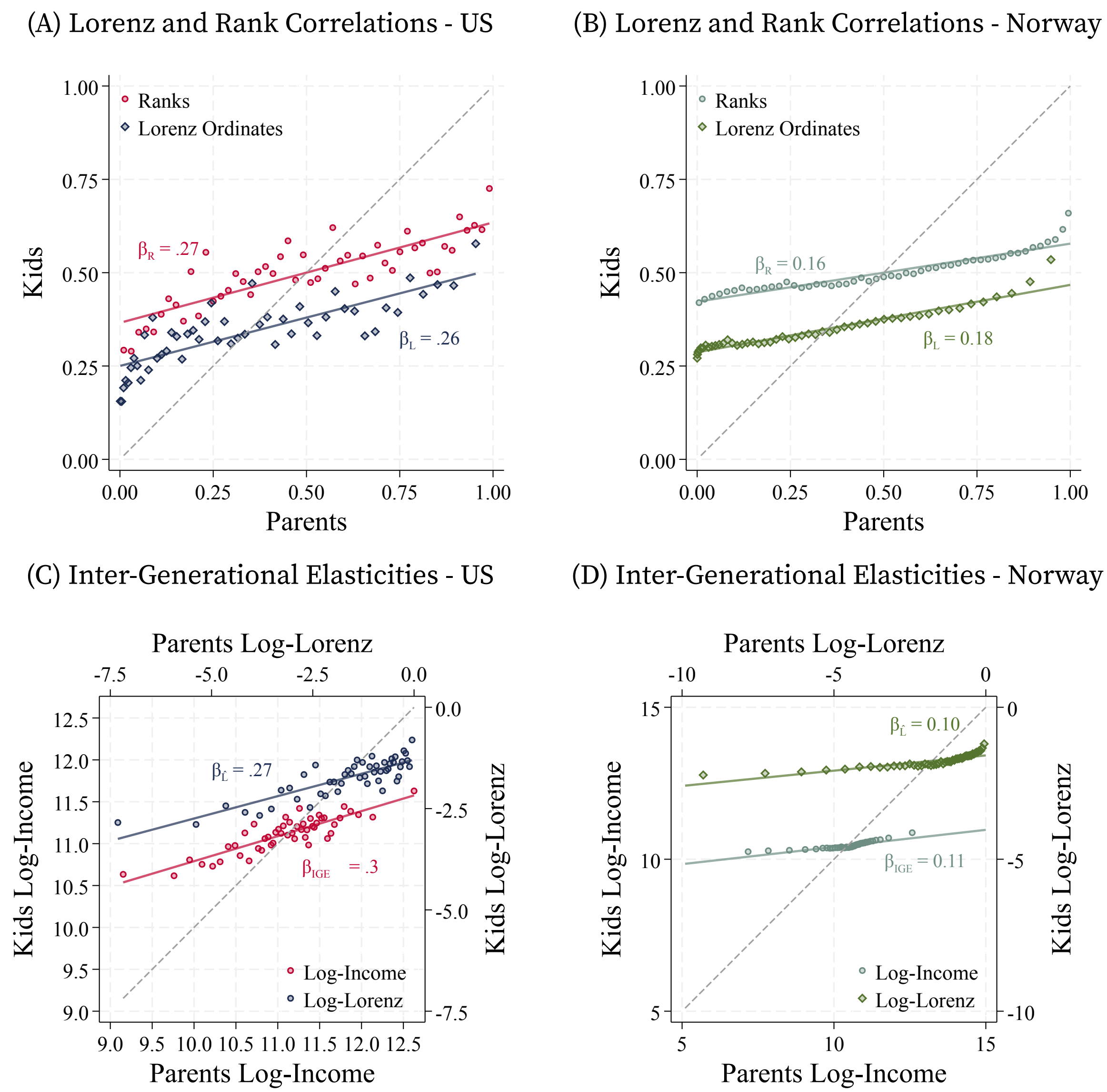
We constrain how status depends on income by imposing 4 conditions expressed in Axioms 1–4 below. We start by normalizing the scale of status, forcing it to be bounded.

Axiom 1 (**Status Normalization**). *Status satisfies* $s(y, Y) \in [0, 1]$.

We restrict attention to status functions that are strictly monotone in income and right-continuous. Monotonicity is key to distinguish purely ordinal representations of status, like income ranks, from cardinal representations that reflect income levels. Specifically, status increases with the dynasty's income, holding the rest of the income distribution constant. Right-continuity accounts for ties in the finite economy leading

[21] This corresponds to one-fifth of the gap between the most and least mobile cities in Chetty et al. (2014a) for the US and one-third of the regional differences in Bratberg et al. (2017) for Norway.

FIGURE 5. Intergenerational correlations



Notes: Panels A and B plot children's income ranks (circles) and Lorenz ordinates (diamonds) against the corresponding parental values for the US and Norway, respectively. Panels C and D plot children's log-income and log-Lorenz ordinates against the corresponding parental values, using separate axes for the two measures. Points are binned means, solid lines are linear fits, and dashed lines show equality across generations. The reported β coefficients are the slopes of the corresponding fitted lines.

to jumps in status when income changes so as to break those ties.

AXIOM 2 (**Continuity and Monotonicity**). *Status* $s(y_i, Y)$ *is strictly increasing and right-continuous on* y_i *given* $\{y_j\}$ *for* $j \neq i$.

In order to disentangle economic growth from mobility we focus on status functions that capture *relative*, as opposed to *absolute*, mobility. Therefore, status depends on the shape of Y and the relative position of y in Y, but not on their level.

AXIOM 3 (**Growth Independence**). *Status satisfies* $s(y, Y) = s(\kappa y, \kappa Y)$ *for any* $\kappa > 0$.

Finally, we establish how status responds to the shape of the income distribution. Our objective is that status combines the cardinal and ordinal information reflected in income differences. Specifically, we make status *aspirational*: dynasties aspire to be ahead of others and reach those above. Status increases when the dynasty gets closer (in income) to those above and further away from those at the bottom of the distribution, given their position in the distribution. We strengthen this by imposing that status be invariant to pure redistribution above or below the dynasty. This makes status "*anonymous*:" a dynasty does not care about the identities of others (whose positions can be reshuffled by redistribution) while still caring about their own relative position (unchanged by this redistribution). This property makes the cumulative distribution of income the relevant statistic for determining status.

AXIOM 4 (**Aspirational Status**). *Order dynasties in Y such that* $y_i \leq y_{i+1}$ *for all* $i = 1, \ldots, N$. *Let* $1 < h < j \leq N$ *be two dynasties and make any transfer* δ *between them such that*

$$y_{h-1} < y_h + \delta < y_{j+1} \qquad \textit{and} \qquad y_{h-1} < y_j - \delta < y_{j+1} \tag{16}$$

letting $y_{j+1} = y_j$ *if* $j = N$. *The transfer* δ *can be positive or negative. Then,*

(a) *The status of dynasties* $i = 1, \ldots, h-1, j+1, \ldots, N$ *remains unchanged:* $s(y_i, Y) = s\left(y_i, Y^{\delta}\right)$ *where* Y^{δ} *is the same as* Y *except for the incomes of the* h *and* j *entries; and*

(b) *The status of dynasty* j *is unchanged if their rank is unchanged: if* $y_{j-1} < y_j - \delta < y_{j+1}$, *then* $s\left(y_j, Y\right) = s\left(y_j - \delta, Y^{\delta}\right)$.

Together, these axioms tie status, and hence mobility, to the Lorenz ordinates of dynasties in the income distribution as well as to their income rank. Status reflects cardinal differences in income, because increasing income brings a dynasty closer to those they aspire to reach, even when income ranks remain unchanged.

PROPOSITION 1. *A status function satisfies Axioms 1–4 if and only if*

$$s(y, Y) = \psi(r(y, Y), L(y, Y)), \tag{17}$$

for $\psi : [0, 1] \times [0, 1] \longrightarrow [0, 1]$ *a continuous function increasing in its first argument and strictly increasing in the second.*

PROOF. Let Y be such that $y_i \leq y_{i+1}$ for all $i = 1, \ldots, N$ and fix a dynasty i. Consider an alternative $N \times 1$ ordered income vector Y' with the same average income, $\mu' = \mu$, and cumulative income among those richer than i, $\sum_{\{\tilde{y}' > y'_i\}} \tilde{y}' = \sum_{\{\tilde{y} > y_i\}} \tilde{y}$. Assume the vectors coincide in their i^{th} entry, $y'_i = y_i$, so the sum of income below i also coincides.

It is possible to move from Y to Y' by means of transfers among dynasties $1, \ldots, i-1$ and dynasties $i+1, \ldots, N$. Axiom 4 demands $s(y_i, Y) = s\left(y_i, Y'\right)$. Therefore, status cannot depend on the whole distribution of income but only on y_i, its rank, r, mean income, μ, and the sum of income above y_i, $\sum_{\{\tilde{y} > y_i\}} \tilde{y}$.

A transfer from i to $h < i$, that does not change i's rank does not change status (Axiom 4). So, status cannot depend on y_i directly, except for its effect on the cumulative income sum, $\sum_{\{\tilde{y} \leq y_i\}} \tilde{y}$, which implies income above y_i.

Axiom 3 implies μ is superfluous—re-scaling income results in the same status—so we are left with two arguments (ranks and cumulative income shares) and write $s(y, Y) = \psi\left(r(y, Y), \sum_{\{\tilde{y} \leq y\}} \frac{\tilde{y}}{N\mu}\right)$ for some function ψ. The second argument is the Lorenz ordinate, which, unlike the rank, is strictly increasing in income. Hence, ψ must be strictly increasing in the second argument by Axiom 2. Ranks are only weakly increasing in income, as not all changes in income change relative positions. Thus, ψ depends only weakly on them. Finally, ψ must be continuous to preserve the right-continuity of ranks and Lorenz ordinates per Axiom 1. $\square$

We further isolate how status depends on the shape of the income distribution by restricting how it moves across co-monotone distributions. Specifically, we require status moves linearly along the ray connecting any two such distributions. This condition sets how status units deal with the cardinality of income, as the position (rank) of each dynasty is fixed along these movements.

Axiom 5 (**Co-Monotone Rays**). *Order dynasties in Y such that $y_i \leq y_{i+1}$ for all $i = 1, \ldots, N$. Consider an alternative income vector Y' with the same strict order for all dynasties. An economy along the ray connecting these two vectors has $Y^\alpha = \alpha Y + (1-\alpha) Y'$, for $\alpha \in [0,1]$. Status satisfies $s\left(y_i^\alpha, Y^\alpha\right) = \alpha s\left(y_i, Y\right) + (1-\alpha) s\left(y_i', Y'\right)$.*

Axiom 5 only covers mixing of co-monotone income distributions. The outcome of transfers that change the rank of a dynasty is not covered, neither are distributions with rank reversals between dynasties. Among co-monotone distributions, we impose no restrictions on the shape of Lorenz curves or the change in status levels between Y and Y'. In particular, Lorenz curves can cross allowing $s\left(y_i, Y\right) > s\left(y_i', Y'\right)$ for some i and $s\left(y_j, Y\right) < s\left(y_j', Y'\right)$ for some $j \neq i$. Nevertheless, Axiom 5 is enough to obtain that the ψ function in (17) is affine in Lorenz ordinates.

Proposition 2. *A status function satisfies Axioms 1–5 if and only if*

$$s\left(y_i, Y\right) = \Omega\left(r_i\right) \cdot \left[\lambda L_i + (1-\lambda)\Xi\left(r_i\right)\right], \tag{18}$$

for weakly increasing functions $\Omega(r) \in (0,1]$ and $\Xi(r) \in [0,1]$ and scalar $\lambda \in (0,1]$, where $r_i = r\left(y_i, Y\right)$ and $L_i = L\left(y_i, Y\right)$ are the rank and Lorenz ordinate of dynasty i.

Proof. Let Y such that $y_i \leq y_{i+1}$ with strict inequality for at least one i and Y' a income with the same strict order. Without loss, we have $\mu = \mu'$, as status and the Lorenz curves are scale invariant (Axiom 3). Under Proposition 1, Axiom 5 requires

$$\begin{aligned} s\left(y_i^\alpha, Y^\alpha\right) &= \psi\left(r\left(y_i^\alpha, Y^\alpha\right), L\left(y_i^\alpha, Y^\alpha\right)\right) \\ &= \alpha\psi\left(r\left(y_i, Y\right), L\left(y_i, Y\right)\right) + (1-\alpha)\psi\left(r\left(y_i', Y'\right), L\left(y_i', Y'\right)\right). \end{aligned} \tag{19}$$

From Aaberge (2001), the Lorenz is linear under co-monotone mixtures,

$$L\left(y_i^\alpha, Y^\alpha\right) = \alpha L\left(y_i, Y\right) + (1-\alpha) L\left(y_i', Y'\right). \tag{20}$$

Hence, ψ satisfies (19) if and only if it is affine on L, with coefficients that depend weakly on ranks, giving (18). The coefficient on L must be greater than zero to ensure strict monotonicity of status on income (Axiom 2). If Y and Y' are such that $y_i = y_j$ and $y_i' = y_j'$ for all i, j, Axiom 3 implies $s\left(y, Y\right) = s\left(y', Y'\right) = s\left(y^\alpha, Y^\alpha\right)$, which satisfies Axiom 5 immediately. Axiom 1 restricts ranges to ensure $s \in [0,1]$. □

Thus, status depends on the Lorenz ordinates of dynasties, their position in the Lorenz curve, and not just in their ranks, their position in the income distribution. This endows status with cardinal information, sets the units and range of status, and makes explicit the link between inequality and status (as discussed in Section 2). The (increasing) weights on the Lorenz ordinates can capture social preferences that depend on the position of individuals, reinforcing the role of inequality.[22] Finally, while our representation of status goes beyond ranks, they are a limiting case when $\lambda \longrightarrow 0$, $\Omega(r) = 1$, and $\Xi(r) = r$. When $\lambda = 1$ and $\Omega(r) = 1$ status coincides with Lorenz ordinates.

6. Conclusions

Our approach to measuring mobility, by adopting Lorenz ordinates, have several advantages for applied research. It directly addresses the concern that purely ordinal measures, capturing rank displacement, may overstate mobility by dispensing with the economic significance of changes. This brings standard concepts from the study of inequality to the study of mobility and incorporates both ordinal and cardinal information. In sufficiently equal societies, Lorenz ordinates are close to rank-based measures of status. In unequal societies, differences in Lorenz ordinates are small where the income distribution is compressed and large where income is dispersed, capturing material differences between individuals.

We see this when applying this framework to intergenerational mobility in the US and Norway we find Lorenz mobility is consistently lower than rank mobility, and, in the case of signed-mobility, often disagree in direction. This highlights the role of inequality (and its changes) in the measurement of mobility. The same results arise when measuring standard intergenerational persistent measures, with the correlation

[22]This is consistent with rank based weights in Yaari's 1988 evaluation measure, the parametric weighting in Donaldson and Weymark's 1980 S-Gini index, as well as generalized concentration indexes (Kakwani, Wagstaff, and Van Doorslaer 1997,Wagstaff 2002).

of ranks being lower than that of Lorenz ordinates, which is in turn lower than the correlation of (log) income. However, taken together, these measures of intergenerational correlation produce similar evaluations of economic mobility. This is in part because Lorenz ordinates have similar concordance as ranks and income, which produces a common ordering over mobility measures (McGee 2025).

Additionally, the link between inequality and mobility is complete in the special case of net mobility, which is equivalent to changes in the Gini coefficient. This makes our proposed measure panel independent—allowing mobility to be computed from repeated cross sections. Moreover, the Lorenz curve has a long history of statistical interest and many techniques exist to reconstruct the Lorenz curve from coarse groups. These advantages make a Lorenz based approach particularly attractive in settings where data collection is challenging. Relatedly, the Lorenz curve effectively downweights the impact of measurement error across much of the distribution. As with ranks, measurement error is more important where the density is densest, but the properties of the Lorenz curve make measurement error less important outside of the right tail—typically, a thin region of the income distribution.

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Appendix A. Mobility Measures

We provide standard characterization of status mobility taking the measurement of economic status as given. First, we impose mobility is symmetric, thus it measures the absolute change in status, similar to Fields and Ok (1996) for income levels.

PROPOSITION A1 (**Absolute Status Mobility**). *A mobility function* $m : \mathbb{R}_+ \times \mathbb{R}_+ \longrightarrow \mathbb{R}_+$ *is*

(a) *Continuous in all arguments;*

(b) *Symmetric in that* $m\left(s, s'\right) = m\left(s', s\right)$*;*

(c) *Step-additive in that* $m\left(s, s''\right) = m\left(s, s'\right) + m\left(s', s''\right)$ *for all* $s \leq s' \leq s''$*; and*

(d) *Translation-invariant in that* $m\left(s + \zeta, s' + \zeta\right) = m\left(s, s'\right)$ *for all* s*,* s' *and* $\zeta > 0$*;*

if and only

$$m\left(s^P, s^K\right) \propto \left|s^K - s^P\right| . \tag{A.1}$$

PROOF. Consider status s. By translation invariance $m(s, s + \zeta) = m(0, \zeta)$ for $\zeta \geq 0$. So, mobility does not depend on s, only on the gap ζ. Define mobility by a gap ζ as $g(\zeta) \equiv m(0, \zeta)$. g is immediately continuous because m is continuous. Moreover, mobility is additive on the gap, that is, $g\left(\zeta + \zeta'\right) = g(\zeta) + g\left(\zeta'\right)$. To see this let $\zeta \geq \zeta' \geq 0$, step-additivity of m gives

$$g\left(\zeta + \zeta'\right) = m\left(0, \zeta + \zeta'\right) = m(0, \zeta) + m\left(\zeta, \zeta + \zeta'\right) = g(\zeta) + g\left(\zeta'\right) . \tag{A.2}$$

Therefore, $g(\zeta) = \kappa\,\zeta$ for some $\kappa \geq 0$, because continuous additive functions are linear.

Consider s and s' with $s \leq s'$. We have $m\left(s, s'\right) = g\left(s' - s\right) = \kappa(s' - s)$. Symmetry extends the characterization to all status pairs s and s' with $m\left(s, s'\right) = \kappa\,|s' - s|$.

□

Second, we replace symmetry in favor of *signed-symmetry*.

PROPOSITION A2 (**Signed Status Mobility**). *A mobility function* $m : \mathbb{R} \times \mathbb{R} \longrightarrow \mathbb{R}$ *satisfies properties (a), (c), and (d) of Proposition A1 and is signed-symmetric, in that* $m\left(s, s'\right) = -m\left(s', s\right)$ *with* $m\left(s, s'\right) > 0$ *for all* $s < s'$*, if and only if*

$$\widetilde{m}\left(s^P, s^K\right) \propto s^K - s^P . \tag{A.3}$$

PROOF. Consider status s and s' with $s < s'$. We have $m\left(s, s'\right) = g\left(s' - s\right) = \kappa(s' - s) > 0$ from the proof of Proposition A1. Signed-symmetry extends the characterization to all status pairs s and s' with $m\left(s, s'\right) = \kappa\left(s' - s\right)$ and $\kappa > 0$. □

We adopt the Kolmogorov-Nagumo-de Finetti's theorem axiomatic characterization of *quasi-arithmetic* aggregators (Hardy, Littlewood, and Pólya 1952, Thm. 215).

THEOREM A1 (**Kolmogorov-Nagumo-de Finetti**). *A family of aggregators* $\mathcal{M}_N : \mathbb{R}^N \longrightarrow \mathbb{R}$*, indexed by* N*, is*

(a) *Continuous and strictly increasing in all arguments;*

(b) *Symmetric in that* $\mathcal{M}_N(PM) = \mathcal{M}_N(M)$ *for any permutation matrix* P*;*

(c) *Reflexive in that* $\mathcal{M}_N(m\mathbb{1}_N) = m$ *for any* m*; and*

(d) *Associative in that* $\mathcal{M}_N(M) = \mathcal{M}_N([\bar{m}, \ldots, \bar{m}, m_{K+1}, \ldots, m_N])$*, where* $\bar{m} = \mathcal{M}_K([m_1, \ldots, m_K])$*;*

if and only if there is a continuous function Γ *such that*

$$\mathcal{M}_N(M) = \Gamma^{-1}\left(\frac{1}{N}\sum_{i=1}^{N}\Gamma(m_i)\right). \tag{A.4}$$

The first condition ensures higher mobility for any dynasty is reflected in higher aggregate mobility. The second condition ensures *anonymity*, so the indexes of dynasties are superfluous. The third condition ensures consistency in that if all dynasties have the same mobility $(m_i = m)$ then aggregate mobility must coincide with this movement $(\mathcal{M}_N = m)$. The fourth and final condition is the most restrictive, but it is necessary for the additivity of the aggregator.

We further pin down the aggregator by requiring $\mathcal{M}$ be homogeneous of degree 1. So, if mobility is re-scaled, aggregate mobility is re-scaled in the same way. The result is standard and we state it without proof (see Hardy, Littlewood, and Pólya 1952, Thm. 84).

LEMMA A1 (**Homogeneous Aggregation**). *Let* $\mathcal{M}_N : \mathbb{R}^N \longrightarrow \mathbb{R}$ *satisfy Theorem A1.* $\mathcal{M}_N$ *is homogeneous of degree 1, so that* $\mathcal{M}_N(\kappa M) = \kappa\mathcal{M}_N(M)$ *for all* $\kappa > 0$*, if and only if* $\Gamma(m) = m^\gamma$ *for some* $\gamma \neq 0$*, with the* $\gamma = 0$ *case corresponding to the geometric average.*

When M includes negative values we instead get the equivalent signed power mean with $\Gamma(m) = \text{sign}(m)\,|m|^\gamma$:

$$\mathcal{M}_N(M) = \text{sign}\left(\frac{1}{N}\sum_{i=1}^{N}\text{sign}(m_i)\,|m_i|^\gamma\right) \times \left|\frac{1}{N}\sum_{i=1}^{N}\text{sign}(m_i)\,|m_i|^\gamma\right|^{\frac{1}{\gamma}}. \tag{A.5}$$

Absolute mobility is equal to absolute net mobility, adjusted by either (absolute) downward or upward Lorenz mobility depending on whether overall inequality has increased or decreased.

COROLLARY A1 (**Absolute Mobility**). *Under dynastic mobility (5), average mobility satisfies:*

$$\frac{1}{N}\sum_{i=1}^{N} m_i^a = \underbrace{\frac{1}{2}\left|\mathcal{G}\left(Y^P\right) - \mathcal{G}\left(Y^K\right)\right|}_{\text{Absolute Net Mobility}} + 2\left(\underbrace{\mathcal{D}^s \mathbb{1}_{\{\mathcal{G}(Y^P)\geq\mathcal{G}(Y^K)\}}}_{\text{Downward Lorenz Mobility}} + \underbrace{\mathcal{U}^s \mathbb{1}_{\{\mathcal{G}(Y^P)<\mathcal{G}(Y^K)\}}}_{\text{Upward Lorenz Mobility}}\right)$$

where $\mathcal{G}(Y) = 1 - \frac{2}{N}\sum_{i=1}^{N} L_i$ *is the Gini coefficient and* $\mathcal{U}^s$ *and* $\mathcal{D}^s$ *evaluate upward and downward signed mobility.*

The first term captures net vertical mobility due to overall changes in inequality. The second and third terms in the decomposition in Corollary A1 isolate the changes in Lorenz ordinates that do not come from the overall change in inequality. These changes come instead from the dynasties' positional changes. Take, for instance, a decrease in inequality that reduces the Gini coefficient (a net increase in Lorenz ordinates) and implies positive net mobility (captured in the first term). Absolute mobility accounts for the change in Lorenz ordinate of dynasties with upward as well as downward mobility. Total upward mobility must be larger than downward mobility as it captures both the decrease in inequality and reshuffling from downward mobile dynasties. It is this reshuffling that the decomposition isolates, counting it twice reflecting that downward moves necessarily correspond to an upward move for another dynasty when net mobility is positive.

When aggregating economic status, defined as in Proposition 2, we obtain the following characterization of generalized net mobility:

LEMMA A2 (**Generalized Net Mobility**). *Under Lemma A1 with* $\gamma = 1$*, and Propositions 2 and A2, aggregate mobility has the form:*

$$\widetilde{\mathcal{M}}_N(M) \propto \widetilde{G}\left(Y^P\right) - \widetilde{G}\left(Y^K\right), \tag{A.6}$$

where $\widetilde{G}(Y) \equiv 1 - \frac{2}{N}\sum_{i=1}^{N}\Omega\left(r_i\right) L_i$ *is Weymark's (1981) Generalized-Gini coefficient. When* $\Omega(r) = 1$ *for all* r *we obtain the usual Gini coefficient* $G(Y) = 1 - \frac{2}{N}\sum_{i=1}^{N} L_i$.

A.1. Bounds on absolute mobility

We consider the maximum mobility, $\mathcal{M}$, when $\gamma = 1$ given two income distributions. This corresponds to the value of an assignment from generation P to generation K. Order dynasties in Y^G such that $y_i^G \leq y_{i+1}^G$ for all $i = 1, \ldots, N$ and $G \in \{P, K\}$, the index i determines the order of income and not the identity of the dynasty. Denote by s_i^G the corresponding status of the dynasty with income position i in generation G, we know that $s_i^G \geq 0$ is increasing in i. The assignment (or transport) that maximizes mobility solves

$$\max_T \frac{1}{N}\sum_{i=1}^{N} \left| s^K_{T(i)} - s^P_i \right| \tag{A.7}$$

The optimal assignment is counter-monotone because pair-wise mobility is represented by a Monge matrix M (a sub-modular discrete function) with typical entry $m_{ij} = \left| s^K_j - s^P_i \right|$, so that $T^\star(i) = N + 1 - i$. This result is a direct application of the rearrangement of linear sums in Burkard, Dell'Amico, and Martello (2012, Prop 5.7). The key step requires establishing that M satisfies the Monge property, that for $1 \leq i < k \leq n$ and $1 \leq j < \ell \leq n$ the entries of M satisfy

$$m_{ij} + m_{k\ell} \leq m_{i\ell} + m_{kj} \; ; \tag{A.8}$$

$$\left| s^K_j - s^P_i \right| + \left| s^K_\ell - s^P_k \right| \leq \left| s^K_\ell - s^P_i \right| + \left| s^K_j - s^P_k \right| \; . \tag{A.9}$$

To prove this, consider $i < k$ and define

$$f(x) \equiv \left| x - s^P_i \right| - \left| x - s^P_k \right| = \begin{cases} s^P_i - s^P_k, & x \leq s^P_i \,, \\ 2x - (s^P_i + s^P_k), & s^P_i \leq x \leq s^P_k \,, \\ s^P_k - s^P_i, & x \geq s^P_k \,. \end{cases} \tag{A.10}$$

hence f is non-decreasing in x. We know $s^K_j \leq s^K_\ell$ because $j < \ell$, so $f\left(s^K_j\right) \leq f\left(s^K_\ell\right)$:

$$\left| s^K_j - s^P_i \right| - \left| s^K_j - s^P_k \right| \leq \left| s^K_\ell - s^P_i \right| - \left| s^K_\ell - s^P_k \right| . \tag{A.11}$$

Rearranging yields the inequality in (A.9), following from the sub-modularity of m.

To prove that the counter-monotone assignment is optimal, consider any candidate assignment T that is not strictly decreasing, so that is $i < k$ with $T(i) < T(k)$. Swap the assignment of i and k, leading to a new assignment with $\widehat{T}(i) = T(k)$, $\widehat{T}(k) = T(i)$, and $\widehat{T}(\omega) = T(j)$ for $\omega \neq i, k$. Then, set $j = T(i)$ and $\ell = T(k)$ and apply (A.9) to get that

$$m_{i,T(i)} + m_{k,T(k)} \leq m_{i,\widehat{T}(i)} + m_{k,\widehat{T}(k)} \, . \tag{A.12}$$

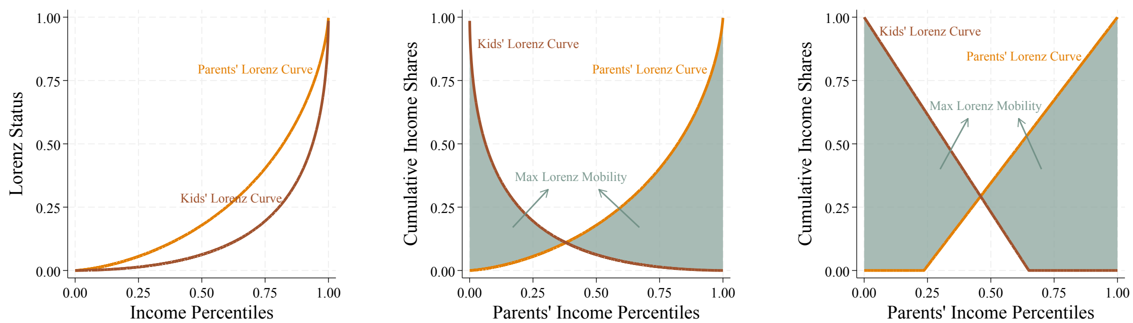
So, the new assignment weakly improves mobility. Save from ties in status, only the counter-monotone assignment is strictly decreasing and maximizes mobility.

Horizontal mobility is also maximized by the counter-monotone assignment as it consist of the special case when the two income distributions coincide and $s^K_i = s^P_i$.

The upper envelope of mobility is obtained by maximising over Lorenz curves. For

FIGURE A.1. Maximal Lorenz Mobility

(A) Initial Lorenz Curves (B) Counter-Monotone Ranks (C) Extreme Lorenz Curves



Notes: Panel A shows the parental and child Lorenz curves. Holding these curves fixed, Panel B pairs parental and child ranks counter-monotonically, which maximizes average absolute Lorenz mobility. Panel C shows the extremal curves used to maximize mobility over Lorenz curves; they have the form $\tilde{L}_c(r) = {}^{r-c}/_{1-c}$ for $r > c$ and 0 otherwise. The shaded areas represent aggregate absolute mobility under $\gamma = 1$.

analytic convenience we compute this in the limiting case as $N \longrightarrow \infty$.[1] Baillo, Carcamo, and Mora-Corral (2022) show that the extreme points in Lorenz space are piecewise linear functions. As the objective function is linear in each curve and we maximise over a convex set, it is sufficient to search for the kink in the two-segment Lorenz curves maximising mobility.[2] That is Lorenz curves defined by kink $c^G \in [0, 1]$ so that $\tilde{L}_{c^G}(r) = 0$ for $r \in [0, c]$, and $\tilde{L}_{c^G}(r) = \frac{r-c^G}{1-c^G}$ for $r \in (c, 1]$.

Mobility is therefore bounded by the maximum mobility under the counter-monotone assignment, integrating $\tilde{m}(r) = \left| \tilde{L}_{c^K}(1-r) - \tilde{L}_{c^P}(r) \right|$,

$$\max_{c^P, c^K \in [0,1]} \int_0^1 \left| \tilde{L}_{c^K}(1-r) - \tilde{L}_{c^P}(r) \right| dr \, . \tag{A.13}$$

There are two cases determined by the breaks in (A.13) at $r = c^P$ and $r = 1 - c^K$:

(a) $0 \leq c^P < 1 - c^K \leq 1$: There are four intervals to consider. One of the curves being

[1] The counter-monotone assignment can be now expressed in terms of ranks, so that $T^\star(r) = 1 - r$. The argument is the same as above but builds directly in the sub-modularity of the difference in status, now defined as functions of rank, r.

[2] Formally, Baillo, Carcamo, and Mora-Corral show that this is the extremal curve for any given value of the Gini coefficient. Searching over kink-points then amounts to searching over Gini coefficients.

zero in two intervals the other two are determined by their crossing at $r = \frac{1-c^K}{2-c^P-c^K}$.

$$\tilde{m}(r) = \begin{cases} \frac{1-c^K-r}{1-c^K} & r \in \left[0, c^P\right] ; \\ \frac{1-c^K-r}{1-c^K} - \frac{r-c^P}{1-c^P} & r \in \left(c^P, \frac{1-c^K}{2-c^P-c^K}\right] ; \\ \frac{r-c^P}{1-c^P} - \frac{1-c^K-r}{1-c^K} & r \in \left(\frac{1-c^K}{2-c^P-c^K}, 1-c^K\right) ; \\ \frac{r-c^P}{1-c^P} & r \in \left[1-c^K, 1\right] . \end{cases} \tag{A.14}$$

The integral is then

$$\int_0^1 \tilde{m}(r)\, dr = \frac{1+c^P}{2} - \frac{\left(c^P\right)^2}{2\left(1-c^K\right)} + \frac{\left(1-c^P-c^K\right)^2\left(\left(1-c^P\right)^2+\left(1-c^K\right)^2\right)}{2\left(1-c^K\right)\left(1-c^P\right)\left(2-c^P-c^K\right)} - \frac{\left(1-c^P-c^K\right)^2}{2\left(1-c^P\right)} . \tag{A.15}$$

(b) $0 \leq 1-c^K \leq c^P \leq 1$: There is a gap between $1-c^K$ and c^P where both functions are zero. So the integral is

$$\int_0^1 \tilde{m}(r)\, dr = \int_0^{1-c^K} 1 - \frac{r}{1-c^K}\, dr + \int_{c^P}^1 \frac{r-c^P}{1-c^P}\, dr = \frac{1-c^K}{2} + \frac{1-c^P}{2} . \tag{A.16}$$

When $1-c^K = c^P$ the lines touch but do not intersect and the integral is equal to 1/2. We can now establish the maximal mobility by selecting the values of c^P and c^P. Crucially, mobility does not depend on the values of c^P and c^P independently, but on the gap between these values and 1: $g \equiv 1-c^P-c^K$. The first case above corresponds to $g > 0$ and the second case to $g \leq 0$. Then, mobility is

$$\int_0^1 \tilde{m}(r)\, dr = \begin{cases} \frac{1+2g-g^2}{2(1+g)} & g > 0 ; \\ \frac{1+g}{2} & g \leq 0 . \end{cases} \tag{A.17}$$

The second case is maximized when $g = 0$ and mobility is 1/2. The critical values of the first case are the roots of $1-2g-g^2$. The only admissible root is $g^\star = \sqrt{2}-1$ which delivers a value of $2-\sqrt{2} \approx 0.59$.

Therefore, the maximal mobility is attained by Lorenz curves of the form $\tilde{L}_{c^G}$ under counter-monotone assignments between generations such that $c^P + c^k = 2-\sqrt{2}$. This tight upper bound implies that

$$0 \leq \mathcal{M}(M) \leq 2 - \sqrt{2} . \tag{A.18}$$

Moreover, this is also the upper bound for horizontal mobility, equivalent to constraining $c^P = c^K = c$ and setting $c = 1 - \frac{1}{\sqrt{2}}$.

A.2. Progressive taxation

LEMMA A3 (**Progressive Taxation and Net Mobility**). *Under Corollary 1, assume Pareto distributed income in each generation $G \in \{P, K\}$, with tail parameter $\alpha^G > 1$, and disposable income satisfying the log-linear tax schedule, with progressivity parameter $\tau^G \in [0, 1)$. Then, before-tax and after-tax net mobility satisfy*

(a) *Before-tax net mobility is positive if and only if $\frac{\alpha^K}{\alpha^P} > 1$ and after-tax net mobility is positive if and only if $\frac{\alpha^K}{\alpha^P} > \frac{1-\tau^K}{1-\tau^P}$.*

(b) *Increasing progressivity in the children's generation increases net mobility, while increasing progressivity in the parents' generation decreases it.*

(c) *If progressivity is common across generations, $\tau^P = \tau^K = \tau$, then taxation preserves the direction of net mobility and attenuates its magnitude toward zero.*

PROOF. From Corollary 1, net mobility before and after taxes satisfy, respectively

$$\mathcal{M}_N\left(M^s\right) \propto \frac{1}{2\alpha^P - 1} - \frac{1}{2\alpha^K - 1} = \frac{2\left(\alpha^K - \alpha^P\right)}{\left(2\alpha^P - 1\right)\left(2\alpha^K - 1\right)}, \tag{A.19}$$

$$\widehat{\mathcal{M}}_N\left(M^s\right) \propto \frac{1 - \tau^P}{2\alpha^P + \tau^P - 1} - \frac{1 - \tau^K}{2\alpha^K + \tau^K - 1}. \tag{A.20}$$

These expressions immediately give $\mathcal{M}_N\left(M^s\right) > 0$ if and only if $\frac{\alpha^K}{\alpha^P} > 1$, and $\widehat{\mathcal{M}}_N\left(M^s\right) > 0$ if and only if $\frac{\alpha^K}{\alpha^P} > \frac{1-\tau^K}{1-\tau^P}$. Taking derivatives gives the result for the change in after-tax mobility to the progressivity of taxes in each generation:

$$\frac{\partial \widehat{\mathcal{M}}_N\left(M^s\right)}{\partial \tau^K} = \frac{2\alpha^K}{\left(2\alpha^K + \tau^K - 1\right)^2} > 0 \quad \text{and} \quad \frac{\partial \widehat{\mathcal{M}}_N\left(M^s\right)}{\partial \tau^P} = -\frac{2\alpha^P}{\left(2\alpha^P + \tau^P - 1\right)^2} < 0. \tag{A.21}$$

Finally, setting $\tau^P = \tau^K = \tau$ we obtain

$$\widehat{\mathcal{M}}_N\left(M^s\right) \propto \frac{2\left(1 - \tau\right)\left(\alpha^K - \alpha^P\right)}{\left(2\alpha^P + \tau - 1\right)\left(2\alpha^K + \tau - 1\right)}, \tag{A.22}$$

$$\frac{\left|\widehat{\mathcal{M}}_N\left(M^s\right)\right|}{\left|\mathcal{M}_N\left(M^s\right)\right|} = \frac{\left(1 - \tau\right)\left(2\alpha^P - 1\right)\left(2\alpha^K - 1\right)}{\left(2\alpha^P + \tau - 1\right)\left(2\alpha^K + \tau - 1\right)} < 1 \tag{A.23}$$

□

The role of progressive taxation in other common mobility measures is stark. Consider the intergenerational elasticities of income and disposable income discussed in footnote 15, respectively,

$$\log\left(y_i^K\right) = \beta_0 + \beta_1 \log\left(y_i^P\right) + u_i \tag{A.24}$$

$$\log\left(y_{i,\text{disp}}^K\right) = \beta_0^{\text{disp}} + \beta_1^{\text{disp}} \log\left(y_{i,\text{disp}}^P\right) + u_{i,\text{disp}}, \tag{A.25}$$

which, under log-linear taxation, yields

$$\beta_1^{\text{disp}} = \beta_1 \frac{(1-\tau_K)}{(1-\tau_P)}. \tag{A.26}$$

Thus, the intergenerational elasticity of disposable income differs from that of market income only by the relative difference in progressivity. When both generations face the same progressive tax system, for example if market income of both generations is measured at the same point in time (as in many data sources which use same time different ages), then the intergenerational elasticity also inherits an irrelevance of progressive taxation.

The following lemma makes the connection of rank dependence and progressive taxation explicit by combining Pareto distributed income as in (13) with the Farlie-Gumbel-Morgenstern (FGM) copula describing the correspondence of ranks between generations. The usefulness of the FGM copula is that it gives a closed-form single-parameter condition in the dependence parameter.

LEMMA A4 (**Progressive Taxation and Absolute Mobility with FGM Dependence**). *Suppose incomes in generation $G \in \{K, P\}$ are Pareto with parameter $\alpha_G > 1$ and tax progressivity $\tau_G \in [0, 1)$. Let*

$$p_G \equiv \frac{\alpha_G}{\alpha_G - 1 + \tau_G}, \tag{A.27}$$

denote the curvature of the after-tax Lorenz curve. If the jointness of the distribution of incomes is the FGM copula,

$$C_\theta(u, v) = uv1 + \theta(1-u)(1-v), \qquad \theta \in [-1, 1], \tag{A.28}$$

then, under aggregator (6) with , $\gamma = 1$, and dynastic mobility (5),

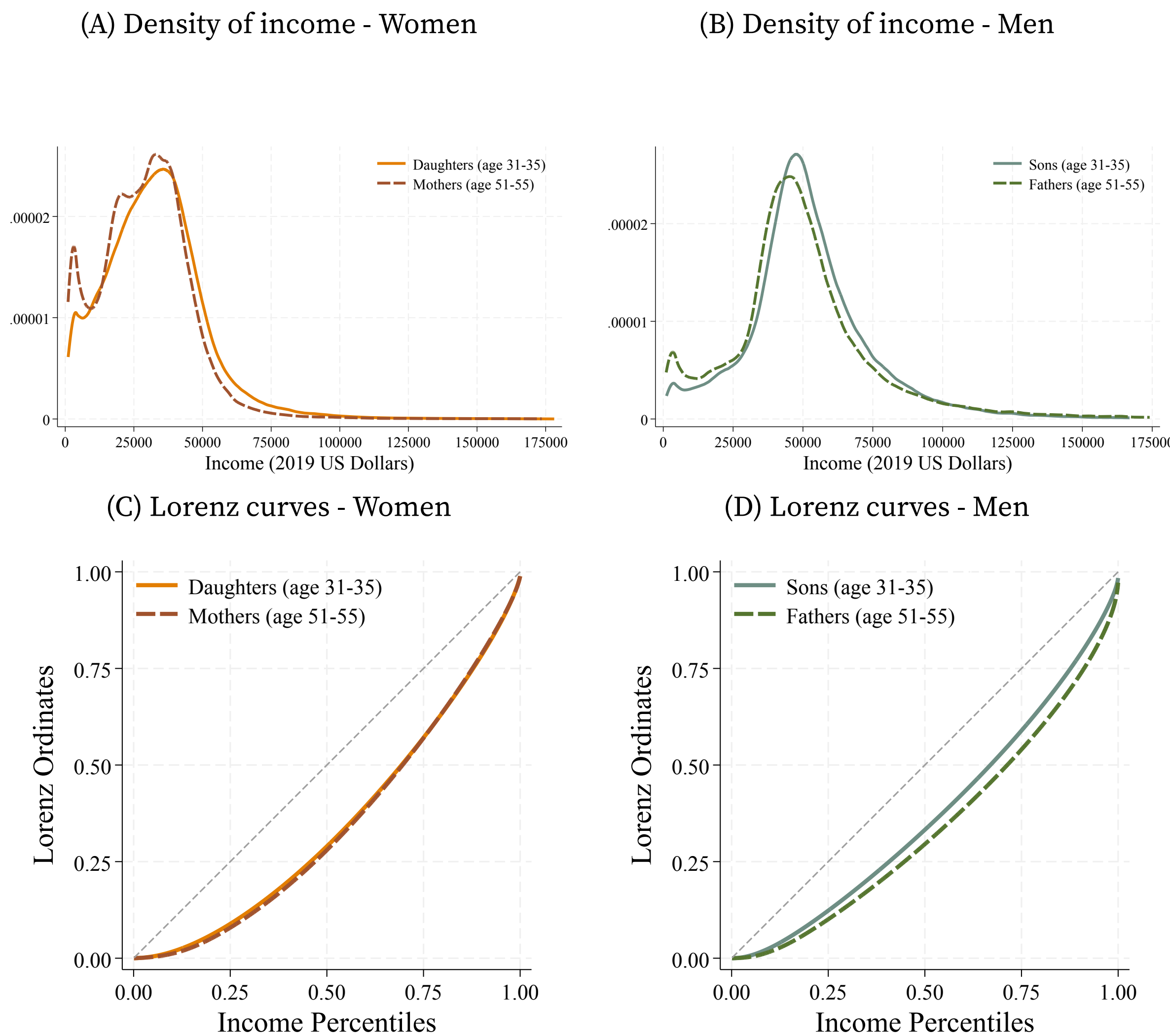
$$\mathcal{M}_N(M^a) \propto \frac{p_K}{p_K+1} + \frac{p_P}{p_P+1} - 2\int_0^1 C_\theta\left(1 - t^{p_K}, 1 - t^{p_P}\right) dt, \tag{A.29}$$

where the effect of (common-)tax progressivity is affine in θ and there exists a positive threshold θ_G^* such that progressivity raises absolute mobility if and only if $\theta < \theta_G^*$.

A.3. Mobility and tax progressivity in Norway: Additional results

We construct two additional samples with the Norwegian data where we distinguish between female and male mobility. First, a sample of mothers and daughters defined by

FIGURE A.2. Income distribution across generations in Norway for women and men



Notes: Panels A and B show kernel density estimates of individual market income for daughters and mothers and for sons and fathers, respectively. Panels C and D show the Lorenz curves for the corresponding income distributions. Income is averaged over ages 31–35 for daughters and sons and ages 51–55 for mothers and fathers and is measured in 2019 US dollars.

the female Norwegians born 1961–1970 with recorded information on their mothers. This sample contains a total of 128,235 paired observations. Second, an equivalent sample for sons and their fathers with a total of 101,529 observations.

The income distributions of daughters and sons move rightwards relative to their mothers and fathers, reflecting overall income growth, and have less mass concentrated at low incomes, as seen in Figures A.2A and A.2B. Daughters' income is 18% higher than their mothers', with the bottom 10 percentiles of the distribution growing more than 70%. Sons' income growth is smaller, slightly below 11%, and less widespread, with the top 15 percentiles of sons actually mapping to lower income levels than their fathers'. After-tax income is markedly less unequal ... what to highlight?

Changes in income distributions across generations are reflected in the Lorenz curves. As seen in Figure A.2C, there are no significant differences between the Lorenz curves of daughters and mothers, despite crossing at the top of the income distribution. In contrast, the Lorenz curve of sons shifts towards equality, Figure A.2D, reflecting the higher income levels of the bottom 10 percentiles and lower income levels of the top 15.

Intergenerational mobility. We find higher mobility in economic status for women than men across most mobility measures presented in Table A.1. As we see from our main mobility measure ($\mathcal{M}$), women's status across generations moves on average by 29% of their aggregate income compared to men's 26%. Status mobility among women corresponds mostly to horizontal mobility, reflecting changes in the position of dynasties, as there are only small differences between the Lorenz curves of mothers and daughters (Figure A.2C). The lack of vertical mobility is further evidenced by the lack of net (or Gini) mobility ($\widetilde{\mathcal{M}}$).

Mobility of economic status is broad-based for women and men alike. We can see this by comparing our main measure of aggregate mobility $\mathcal{M}$ without curvature, $\gamma = 1$, to measures with lower and higher curvature. When we set $\gamma = 1/2$, aggregate mobility favors a more even distribution of mobility, reducing $\mathcal{M}$ if mobility is concentrated among few dynasties. However, mobility remains high, with women's status moving by 23% of total income and men's moving by 22%. Conversely, $\mathcal{M}$ increases when setting $\gamma = 2$, as it weights more dynasties with higher mobility. These results imply that mobility is broad-based, but with some dynasties experiencing larger status changes.

Finally, we look at intergenerational correlations for status, ranks, and log-income levels in the last four columns of Table A.2. As expected, these different measures provide a consistent picture of intergenerational mobility (see McGee 2025). In particular, the correlation of economic status and income ranks are quantitatively similar. This is because the relatively low income inequality in our application results in Lorenz curves that are approximately linear between the 25$^{\text{th}}$ and 75$^{\text{th}}$ percentiles of income.

TABLE A.1. Aggregate mobility in Norway

		Aggregate Mobility $\mathcal{M} = \left(\frac{1}{N}\sum(m_i)^\gamma\right)^{\frac{1}{\gamma}}$					
		Signed Mobility			Absolute Mobility		
		$\gamma = 1$	$\gamma = 1/2$	$\gamma = 2$	$\gamma = 1$	$\gamma = 1/2$	$\gamma = 2$
Market Income							
Women	Lorenz	0.005	0.000	0.053	0.285	0.234	0.365
	Ranks	—	0	0.008	0.297	0.249	0.372
Men	Lorenz	0.033	0.003	0.128	0.261	0.214	0.334
	Ranks	—	0	−0.041	0.283	0.235	0.360
Disposable Income							
Norway	Lorenz	0.031	0.002	0.129	0.277	0.232	0.350
	Ranks	—	0	−0.036	0.298	0.250	0.374
Women	Lorenz	0.018	0.001	0.100	0.291	0.243	0.367
	Ranks	—	0	0.020	0.298	0.250	0.372
Men	Lorenz	0.027	0.002	0.113	0.262	0.218	0.333
	Ranks	—	0	−0.052	0.286	0.238	0.362

Notes: The table reports aggregate signed and absolute mobility in Lorenz ordinates and income ranks for the Norwegian samples. Women denotes mother–daughter pairs, Men denotes father–son pairs, and Norway denotes the pooled parent–child sample. The upper panel uses market income and the lower panel uses disposable income. Signed mobility is the change in a dynasty's status, while absolute mobility is the magnitude of that change. Aggregate signed mobility is computed using the signed power mean in Lemma A1; aggregate absolute mobility uses the standard power mean. The case $\gamma = 1$ is the arithmetic average, $\gamma = 1/2$ places relatively more weight on broadly shared mobility, and $\gamma = 2$ places relatively more weight on larger movements. The signed-rank entries for $\gamma = 1$ are omitted because average signed rank mobility is zero by construction.

TABLE A.2. Intergenerational correlations in Norway

	Intergenerational Correlations $\rho\left(x_i^P, x_i^K\right)$			
	Lorenz	Rank	Log-Lorenz	Log-income
Market Income				
Women	0.181	0.168	0.109	0.108
Men	0.254	0.223	0.124	0.121
Disposable Income				
Norway	0.185	0.163	0.116	0.092
Women	0.174	0.168	0.098	0.102
Men	0.243	0.212	0.181	0.137

Notes: The table reports Pearson correlations between parental and child outcomes in the Norwegian samples. Women denotes mother–daughter pairs, Men denotes father–son pairs, and Norway denotes the pooled parent–child sample. The upper panel uses market income and the lower panel uses disposable income. The Lorenz and Rank columns use Lorenz ordinates and income ranks in levels; the Log-Lorenz and Log-income columns use their logarithms. Higher correlations indicate greater intergenerational persistence and therefore lower mobility.